БИОЛОГИЯ · 10 КЛАСС · УРОК 1

Молекулярная биология и биохимия

Основа: Biology Grade 10, Астана-кітап, 2019. Далее в этом PDF приложены все оригинальные страницы главы, включая рисунки, задания и таблицы. Русский раздел — учебный навигатор и проверочные задания, а не дословный перевод оригинала.

Как заниматься

Изучайте каждый подраздел на оригинальных страницах, используйте словарь терминов в учебнике, затем письменно отвечайте на пять вопросов. После всех подразделов выполните итоговую проверку из 40 вопросов. Проходной порог: 28 правильных ответов.

1.1 Вода

Откройте раздел 1.1 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Вода» и почему это важно.

1. Дайте определение понятия «Вода».
2. Какую роль играет «Вода» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Вода»?
4. Приведите биологический пример, относящийся к теме «Вода».
5. Как можно объяснить или проверить одно из утверждений из раздела «Вода»?

1.2 Простые углеводы

Откройте раздел 1.2 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Простые углеводы» и почему это важно.

1. Дайте определение понятия «Простые углеводы».
2. Какую роль играет «Простые углеводы» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Простые углеводы»?
4. Приведите биологический пример, относящийся к теме «Простые углеводы».
5. Как можно объяснить или проверить одно из утверждений из раздела «Простые углеводы»?

1.3 Полисахариды

Откройте раздел 1.3 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Полисахариды» и почему это важно.

1. Дайте определение понятия «Полисахариды».
2. Какую роль играет «Полисахариды» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Полисахариды»?
4. Приведите биологический пример, относящийся к теме «Полисахариды».
5. Как можно объяснить или проверить одно из утверждений из раздела «Полисахариды»?

1.4 Редуцирующие и нередуцирующие сахара

Откройте раздел 1.4 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Редуцирующие и нередуцирующие сахара» и почему это важно.

1. Дайте определение понятия «Редуцирующие и нередуцирующие сахара».
2. Какую роль играет «Редуцирующие и нередуцирующие сахара» в живых организмах?

3. Какие ключевые структуры, этапы или компоненты связаны с темой «Редуцирующие и нередуцирующие сахара»?

4. Приведите биологический пример, относящийся к теме «Редуцирующие и нередуцирующие сахара».

5. Как можно объяснить или проверить одно из утверждений из раздела «Редуцирующие и нередуцирующие сахара»?

1.5 Липиды

Откройте раздел 1.5 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Липиды» и почему это важно.

1. Дайте определение понятия «Липиды».

2. Какую роль играет «Липиды» в живых организмах?

3. Какие ключевые структуры, этапы или компоненты связаны с темой «Липиды»?

4. Приведите биологический пример, относящийся к теме «Липиды».

5. Как можно объяснить или проверить одно из утверждений из раздела «Липиды»?

1.6 Классификация липидов

Откройте раздел 1.6 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Классификация липидов» и почему это важно.

1. Дайте определение понятия «Классификация липидов».

2. Какую роль играет «Классификация липидов» в живых организмах?

3. Какие ключевые структуры, этапы или компоненты связаны с темой «Классификация липидов»?

4. Приведите биологический пример, относящийся к теме «Классификация липидов».

5. Как можно объяснить или проверить одно из утверждений из раздела «Классификация липидов»?

1.7 Белки

Откройте раздел 1.7 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Белки» и почему это важно.

1. Дайте определение понятия «Белки».

2. Какую роль играет «Белки» в живых организмах?

3. Какие ключевые структуры, этапы или компоненты связаны с темой «Белки»?

4. Приведите биологический пример, относящийся к теме «Белки».

5. Как можно объяснить или проверить одно из утверждений из раздела «Белки»?

1.8 Функции белков

Откройте раздел 1.8 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Функции белков» и почему это важно.

1. Дайте определение понятия «Функции белков».

2. Какую роль играет «Функции белков» в живых организмах?

3. Какие ключевые структуры, этапы или компоненты связаны с темой «Функции белков»?

4. Приведите биологический пример, относящийся к теме «Функции белков».

5. Как можно объяснить или проверить одно из утверждений из раздела «Функции белков»?

1.9 Денатурация белков

Откройте раздел 1.9 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Денатурация белков» и почему это важно.

1. Дайте определение понятия «Денатурация белков».

2. Какую роль играет «Денатурация белков» в живых организмах?

3. Какие ключевые структуры, этапы или компоненты связаны с темой «Денатурация белков»?

4. Приведите биологический пример, относящийся к теме «Денатурация белков».

5. Как можно объяснить или проверить одно из утверждений из раздела «Денатурация белков»?

1.10 Определение белков

Откройте раздел 1.10 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Определение белков» и почему это важно.

1. Дайте определение понятия «Определение белков».

2. Какую роль играет «Определение белков» в живых организмах?

3. Какие ключевые структуры, этапы или компоненты связаны с темой «Определение белков»?

4. Приведите биологический пример, относящийся к теме «Определение белков».

5. Как можно объяснить или проверить одно из утверждений из раздела «Определение белков»?

1.11 ДНК

Откройте раздел 1.11 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «ДНК» и почему это важно.

1. Дайте определение понятия «ДНК».

2. Какую роль играет «ДНК» в живых организмах?

3. Какие ключевые структуры, этапы или компоненты связаны с темой «ДНК»?

4. Приведите биологический пример, относящийся к теме «ДНК».

5. Как можно объяснить или проверить одно из утверждений из раздела «ДНК»?

1.12 Репликация ДНК

Откройте раздел 1.12 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Репликация ДНК» и почему это важно.

1. Дайте определение понятия «Репликация ДНК».

2. Какую роль играет «Репликация ДНК» в живых организмах?

3. Какие ключевые структуры, этапы или компоненты связаны с темой «Репликация ДНК»?

4. Приведите биологический пример, относящийся к теме «Репликация ДНК».

5. Как можно объяснить или проверить одно из утверждений из раздела «Репликация ДНК»?

1.13 РНК

Откройте раздел 1.13 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «РНК» и почему это важно.

1. Дайте определение понятия «РНК».

2. Какую роль играет «РНК» в живых организмах?

3. Какие ключевые структуры, этапы или компоненты связаны с темой «РНК»?

4. Приведите биологический пример, относящийся к теме «РНК».

5. Как можно объяснить или проверить одно из утверждений из раздела «РНК»?

1.14 Нуклеиновые кислоты

Откройте раздел 1.14 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Нуклеиновые кислоты» и почему это

важно.

1. Дайте определение понятия «Нуклеиновые кислоты».
2. Какую роль играет «Нуклеиновые кислоты» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Нуклеиновые кислоты»?
4. Приведите биологический пример, относящийся к теме «Нуклеиновые кислоты».
5. Как можно объяснить или проверить одно из утверждений из раздела «Нуклеиновые кислоты»?

ИТОГОВАЯ ПРОВЕРКА · 40 ВОПРОСОВ

Вопросы открытого типа. Сверяйте ответы с учебником. За каждый полностью правильный ответ — 1 балл; 28/40 — зачёт.

1. Объясните основной механизм, рассматриваемый в теме «Вода».
2. Объясните основной механизм, рассматриваемый в теме «Простые углеводы».
3. Объясните основной механизм, рассматриваемый в теме «Полисахариды».
4. Объясните основной механизм, рассматриваемый в теме «Редуцирующие и нередуцирующие сахара».
5. Объясните основной механизм, рассматриваемый в теме «Липиды».
6. Объясните основной механизм, рассматриваемый в теме «Классификация липидов».
7. Объясните основной механизм, рассматриваемый в теме «Белки».
8. Объясните основной механизм, рассматриваемый в теме «Функции белков».
9. Объясните основной механизм, рассматриваемый в теме «Денатурация белков».
10. Объясните основной механизм, рассматриваемый в теме «Определение белков».
11. Объясните основной механизм, рассматриваемый в теме «ДНК».
12. Объясните основной механизм, рассматриваемый в теме «Репликация ДНК».
13. Объясните основной механизм, рассматриваемый в теме «РНК».
14. Объясните основной механизм, рассматриваемый в теме «Нуклеиновые кислоты».
15. Сравните два явления или компонента, упомянутых в теме «Вода».
16. Сравните два явления или компонента, упомянутых в теме «Простые углеводы».
17. Сравните два явления или компонента, упомянутых в теме «Полисахариды».
18. Сравните два явления или компонента, упомянутых в теме «Редуцирующие и нередуцирующие сахара».
19. Сравните два явления или компонента, упомянутых в теме «Липиды».
20. Сравните два явления или компонента, упомянутых в теме «Классификация липидов».
21. Сравните два явления или компонента, упомянутых в теме «Белки».
22. Сравните два явления или компонента, упомянутых в теме «Функции белков».
23. Сравните два явления или компонента, упомянутых в теме «Денатурация белков».
24. Сравните два явления или компонента, упомянутых в теме «Определение белков».
25. Сравните два явления или компонента, упомянутых в теме «ДНК».
26. Сравните два явления или компонента, упомянутых в теме «Репликация ДНК».
27. Сравните два явления или компонента, упомянутых в теме «РНК».
28. Сравните два явления или компонента, упомянутых в теме «Нуклеиновые кислоты».
29. Как изменение условий повлияет на процесс из темы «Вода»?
30. Как изменение условий повлияет на процесс из темы «Простые углеводы»?
31. Как изменение условий повлияет на процесс из темы «Полисахариды»?
32. Как изменение условий повлияет на процесс из темы «Редуцирующие и нередуцирующие сахара»?
33. Как изменение условий повлияет на процесс из темы «Липиды»?
34. Как изменение условий повлияет на процесс из темы «Классификация липидов»?
35. Как изменение условий повлияет на процесс из темы «Белки»?
36. Как изменение условий повлияет на процесс из темы «Функции белков»?
37. Как изменение условий повлияет на процесс из темы «Денатурация белков»?
38. Как изменение условий повлияет на процесс из темы «Определение белков»?
39. Как изменение условий повлияет на процесс из темы «ДНК»?
40. Как изменение условий повлияет на процесс из темы «Репликация ДНК»?

ОРИГИНАЛЬНАЯ ГЛАВА С ИЛЛЮСТРАЦИЯМИ

Следующие страницы воспроизведены из загруженного пользователем учебника без изменения содержания.

Все учебники Казахстана на OKULYK.KZ

CHAPTER 1.0

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CHAPTER 1

MOLECULAR BIOLOGY AND BIOCHEMISTRY

1. Water
2. Simple carbohydrates
3. Polysaccharides
4. Reducing and Nonreducing sugars
5. Lipids
6. Lipid Classification
7. Protein
8. Protein Function
9. Protein Denaturation
10. Protein Determination
11. DNA
12. DNA Replication
13. RNA
14. Nucleic acids



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1.1 Water

You will:

- explain the fundamental importance of water for life on Earth.

Stimulating question

Why do scientists look for water in other places in the solar system?

Key terms

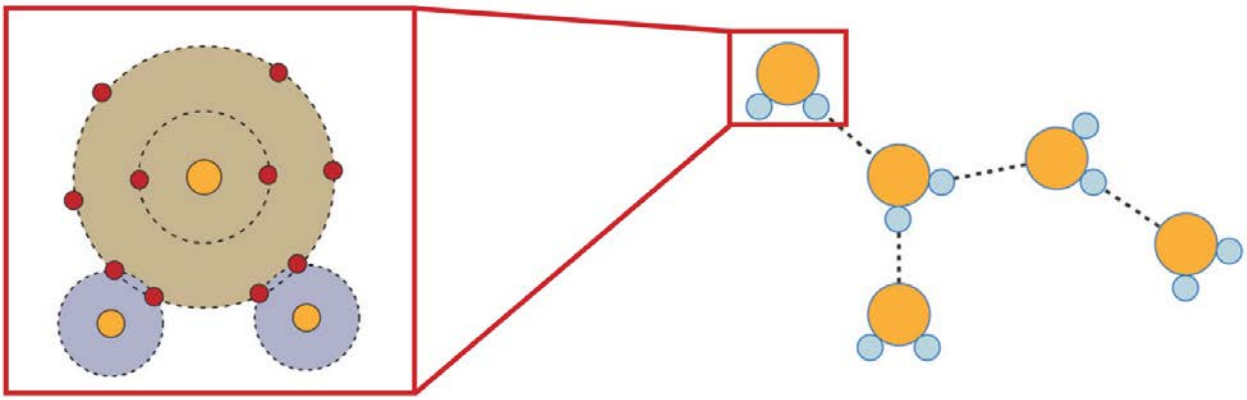
- Cohesion - the sticking together of particles of the same substance;
- Specific heat capacity - the amount of heat per unit mass required to raise the temperature by one degree Celsius;
- Solvent - able to dissolve other substances.

Text

Water is the most abundant component of life. It is an inorganic compound consisting of two hydrogen atoms and one oxygen atom linked by two covalent bonds. Water is a polar molecule, it’s charged partially positively near the hydrogen atoms and partially negatively near the oxygen atom. The negative ends of water molecules attract the positive ends of other water molecules forming hydrogen bonds.

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Water's polarity is caused by its V-shape.

The polarity of water molecules and hydrogen bonding between them give water unique properties which makes it vital to life.

Water is a good solvent. Many substances have molecules with any charged groups. When these substances are added to the solution oxygen attracts positive ions while hydrogen attracts negative ions. Charged particles become dispersed among the water molecules and easily dissolve. Water transports substances as a solution around organisms by transport systems. Metabolic reactions between substances take place in water solutions.

Hydrogen bonds give water molecules cohesive properties. Cohesion is the attraction between water molecules. Adhesion is the attraction of water molecules to a surface. These properties allow water and minerals move upwards through vascular tissue of plants.

A consequence of cohesion is high surface tension, a measure of the difficulty of breaking the surface of a liquid. Water molecules at the surface are exposed to air on one side, and have less water molecules to bond than in deeper within the liquid. They cause water molecules to have stronger bonds at the surface forming an invisible "membrane". Small insects take advantage of surface tension by walking on water.

Another property of water is related to its high specific heat capacity. A specific heat capacity of water is the amount of heat that causes 1g of water to change its temperature by 10C. A large amount of energy is required to raise the temperature of water. When water is heated, some energy added by heating is used to break hydrogen bonds instead of causing more motion among the molecules. The abundant hydrogen bonds make water resistant to temperature changes. This stability is essential for biochemical reactions and for aquatic organisms. It helps maintain a body temperature of mammals and moderates Earth's climate change.

Most liquids increase in density and decrease in volume when become solid.

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However, water has a lower density as a solid than a liquid. This freezing property allows ice to float on water and lakes freeze at the top protecting organisms below.

Activity

1. Create a mind map for water properties and compare with your partner.
2. What would happen if water has a different property? Discuss with your partner and fill the table.

If water	*Consequences*	*Affected organisms*
is a nonpolar molecule		
has extremely low specific heat capacity		
is less dense than ice		
has lower amount of heat to vaporise		

Literacy

1. Which substances cannot dissolve in water and why? Explain your answer.
2. Why ice is less dense than water?
3. Annual change in temperature in cities located near the oceans or seas is much lower than in cities located far from oceans and seas. Explain this phenomenon?

Facts

- Water covers three-quarter of the Earth's surface, but unfortunately, only about 0.3% is usable by humans.
- Water is the only substance found pure in all three states of matter on the Earth.

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Research time

Another important property of water is its transparency. Research and find out its significance for plants and animals. Write a report about it.

Terminology

-
- abundant - мол / обильный;
- consequence - салдар / последствие;
- dispersed - таралған / рассеянный;
- dissolved - еріген / растворенный;
- essential - қажетті / необходимый;
- lattice - тор / решетка;
- motion - қозғалыс / движение;
- resistant - төзімділік / стойкий;
- solvent - еріткіш / растворитель;
- specific heat capacity - меншікті жылу сыйымдылығы / удельная теплоемкость;
- surface tension - беттік керіліс / поверхностное натяжение;
- transparency - мөлдірлік / прозрачность.

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1.2 Simple carbohydrates

You will:

- classify carbohydrates by their structure, composition and functions.

Stimulating question

Where did the carbohydrates get their name from?

Key terms

- Carbohydrates - biological molecules made of carbon, hydrogen, and oxygen atoms;
- Monosaccharide - a simple sugar, the basic unit of carbohydrates;
- Disaccharide - a double sugar composed of two linked molecules of monosaccharides.

Text

Carbohydrates are organic compounds consisting of carbon (C), oxygen (O), and hydrogen (H) atoms. Their general formula is Cn(H2O)n, where n can be equal or more than 3. Carbohydrates are classified into three groups: monosaccharides, disaccharides and polysaccharides.

Monosaccharides

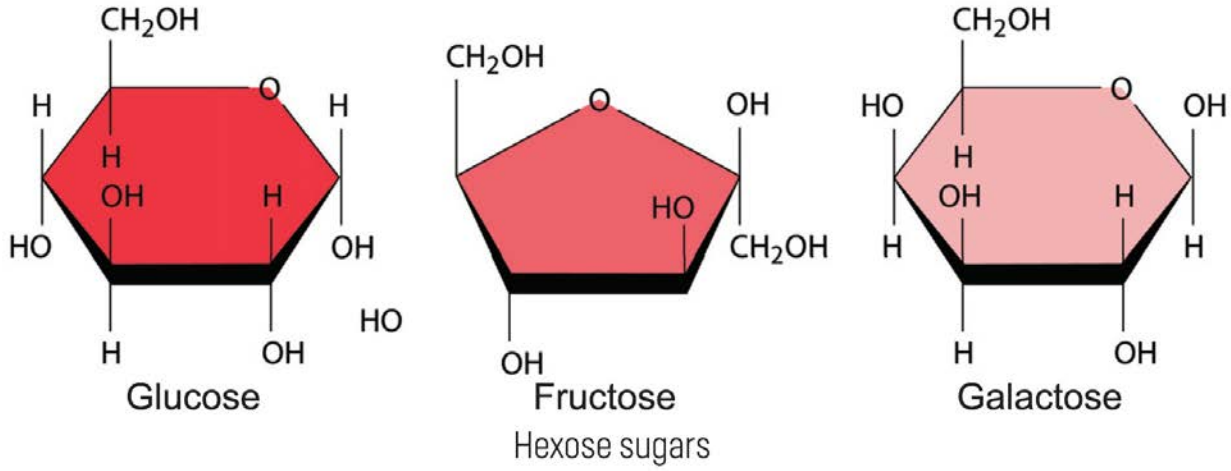
Monosaccharides are simple sugars that usually have three to seven carbon atoms. They can join to make more complex carbohydrates. Their structure contains a carbonyl group (CO) and multiple hydroxyl groups (-OH). According to the position of carbonyl group, they are classified as aldose or ketose. Aldoses’ carbonyl group is located at the end of the carbon skeleton while ketoses’ carbonyl group is located within a carbon skeleton.

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Monosaccharides also can be named according to their number of carbons: some of the most common types are trioses (three C), pentoses (five C), and hexoses (six C). Pentose sugars include deoxyribose and ribose. They are important components of nucleic acids.

Three of the most important hexoses are glucose, fructose, and galactose. They have the same chemical composition ($C_6H_{12}O_6$) but have different characteristics according to the different arrangement of atoms. Such materials are called isomers. Isomers, in general, have different physical properties based on their structure.



Hexose sugars

Glucose is one of the most common monosaccharide used by nearly every form of life. It provides both energy and structure to many organisms. Glucose molecules can be precursors for the cellular respiration to produce energy in the form of ATP. If energy is not needed it bonds with other monosaccharides and stored in the organism.
Fructose has slightly different arrangement of atoms than glucose. It can be produced in organisms from glucose. Fructose is the source of energy, it enhances the absorption of water, sodium, and potassium. Main source of fructose is fruits and honey.
Galactose is produced in many organisms, but mainly in mammals. Galactose is the component of milk; it is used as an energy source by the offspring.

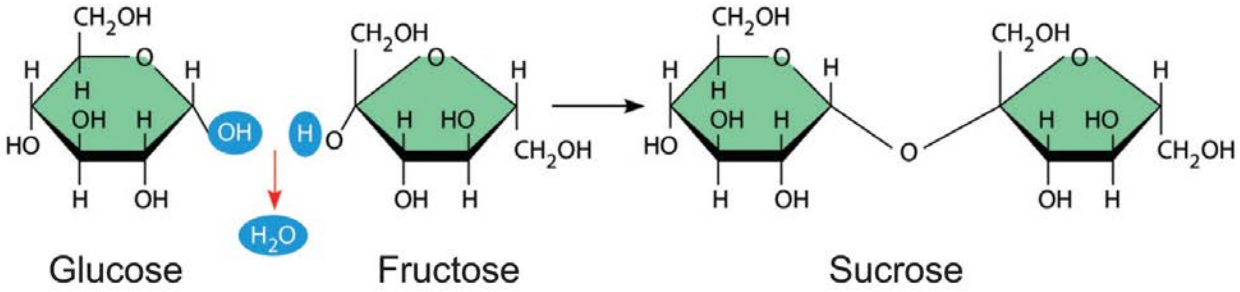
Disaccharides

Disaccharides are crystalline, water-soluble sugars, consisting of two monosaccharides. They are joined by glycosidic bond (linkage). When two monosaccharide molecules react to form a glycosidic linkage, a water molecule

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is generated in the process through a chemical reaction known as dehydration (condensation). The inverse reaction is called hydrolysis. During hydrolysis water is added to break down glycosidic linkages. The three common disaccharides are sucrose, lactose and maltose.



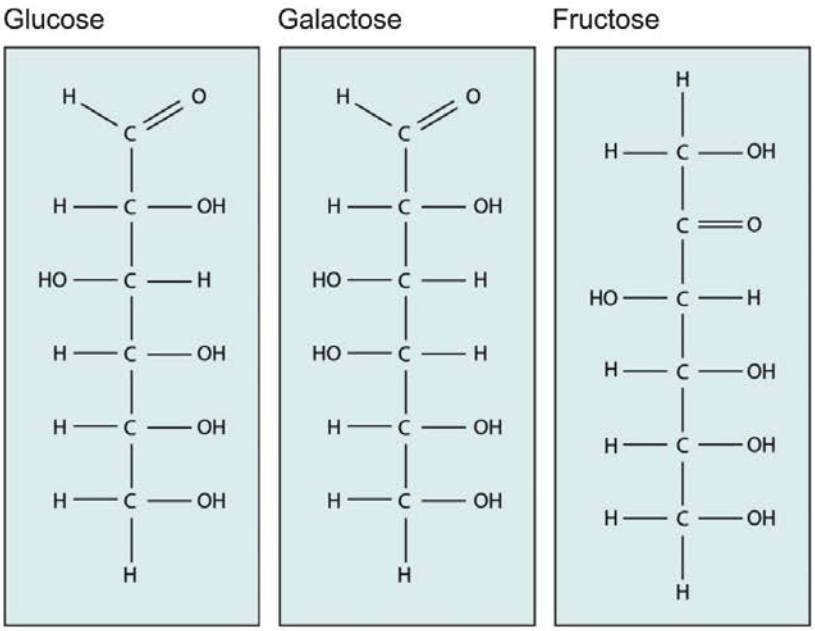
Monosaccharides join by dehydration reaction

Sucrose comes from joining of fructose and glucose. Sucrose is the main component of table sugar. Lactose comes from glucose and galactose joining. It is found in mammal's milk. Maltose is a product of starch breakdown during digestion that consists of two molecules of glucose. It may be purified from barley and other grains.

Activity

1. Label the differences between hexose sugars.
2. Draw the breakdown of lactose and maltose in the chemical reaction.
3. Describe a biological role of the following carbohydrates.

Simple sugars	*Biological role*
Glucose	
Sucrose	
Fructose	



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Facts

Lactose intolerance is an inability to digest and absorb lactose, found in dairy products. It occurs when the small intestine does not make enough lactase enzyme. Symptoms are gases, belly pain, and bloating..

Research time

Find information about pentose sugars. Explain their biological role, and draw their cyclic chemical structures.

Literacy

1. Why carbohydrates are considered as organic molecules?
2. During which process carbohydrates are produced?
3. Describe the processes of dehydration and hydrolysis reactions.

Terminology

- arrangement - орналасу орны / расположение;
- barley - арпа / ячмень;
- dehydration reaction - дегидратация реакциясы / реакция дегидратации;
- glycosidic bond - гликозидтік байланыс / гликозидная связь;
- hydrolysis reaction - гидролиз реакциясы / реакция гидролиза;
- precursor - ізашар, алғызат / предшественник;
- property - қасиет / свойство;
- provide - қамтамасыз ету / обеспечивать;
- to compose - құрастыру / составить;
- to enhance - күшейту / усиливать;
- to generate - шығару / производить;
- to join - біріктіру / соединять;
- to locate - орналастыру / размещать.

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1.3 Polysaccharides

You will:

- classify carbohydrates by their structure, composition and functions.

Stimulating question

What happens to excess glucose in different organisms?

Key terms

- Polysaccharide - a large molecule composed of a long chains of monosaccharide units bound together by glycosidic linkage.
- Glycosidic linkage - is a type of covalent bond, formed between monosaccharides.

Text

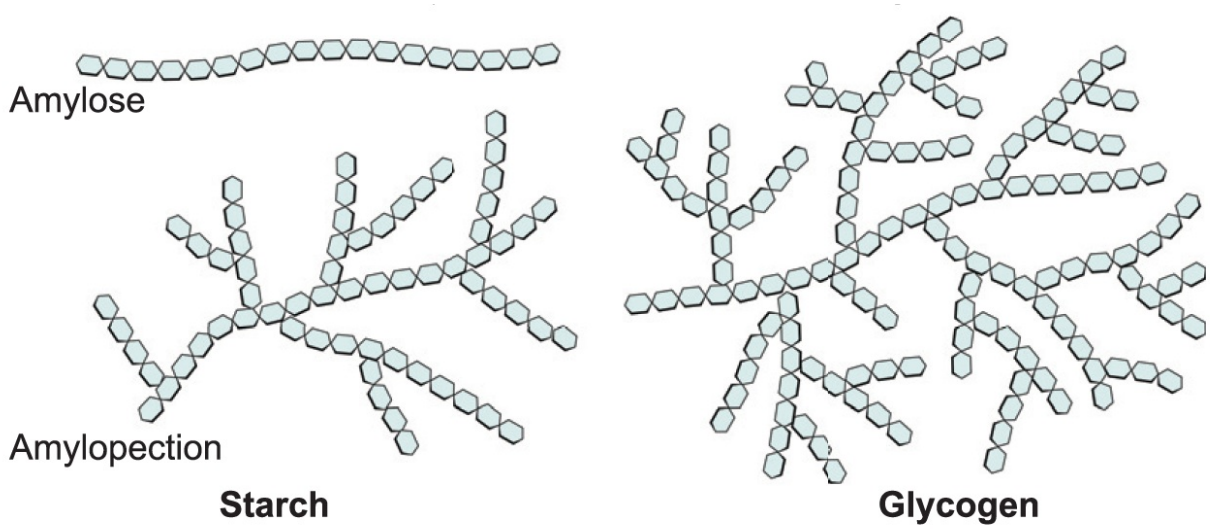
Polysaccharides are complex carbohydrates. They consist of a long chain of monosaccharides linked together by a glycosidic bond. Polysaccharide chain may be branched or unbranched. Their main functions are storage of energy and structural composition of the cell. The major examples of polysaccharides are starch, glycogen, cellulose, and chitin.

Starch is the storage polysaccharide found in plants. It consists of glucose monomers. During photosynthesis, an excess amount of glucose is joined together and is stored in plant organs in the form of starch. Starch has two forms: amylose and amylopectin. Amylose is the linear form of starch where glucose molecules joined by 1-4 linkage (first carbon of glucose joins to fourth carbon of another glucose). It can form a helical structure. This structure of amylose allows cells to store energy in a compact form and makes it easily accessible. Amylopectin is the branched form of starch where glucose molecules joined by 1-6 linkages in branching points. Branching occurs about every 20 glucose units.

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Human, animals, bacteria and fungi store energy in a different form of polysaccharide called glycogen. It is a polymer of glucose, and it is similar to amylopectin but it has larger number of branches. These branches are shorther than those in amylopectin. Branching occurs in every 10 glucose units. Glycogen is generally stored in liver and muscle cells. When organism requires glucose for energy, glycogen is hydrolysed.. Lactose comes from glucose and galactose joining. It is found in mammal's milk. Maltose is a product of starch breakdown during digestion that consists of two molecules of glucose. It may be purified from barley and other grains.



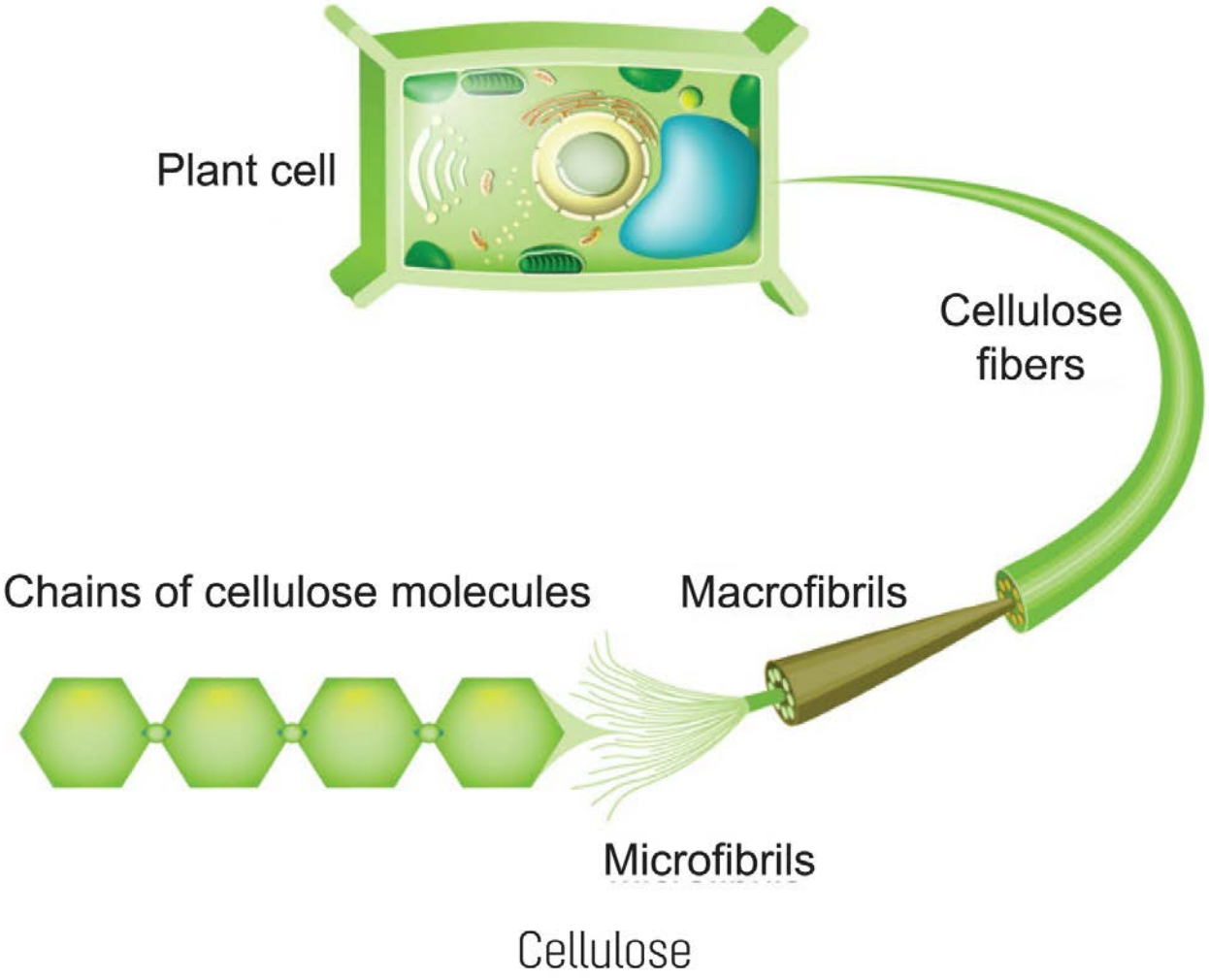
Starch and glycogen

Cellulose is the component of plant cell wall. It has fibrous, linear structure made up of glucose units. Unlike amylose, it has a different glycosidic linkage. When glucose forms cyclic structure, -OH group attached to the first carbon can be either below or above the plane. According to this, glucose is called alpha (α) or beta (β).

In starch, α glucose molecules are located at the same orientation while in cellulose, each β glucose is upside down to the next β glucose. The linear structure of cellulose molecules allows them to bond between -OH groups on closely chains causing a fibre. These bonds give them great mechanical strength.

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Chitin is the structural polysaccharide. It is a component of exoskeleton, found in most arthropods, and cell wall of fungi. Chitin is a polymer of N-glucosamine molecules, derivatives of glucose. These molecules have attached nitrogen-containing group. Like cellulose, monomers of chitin are linked by β-glycosidic bonds.

Activity

Fill the table. Some examples are given. Draw the diagram and shape of polysaccharides in first two columns.

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Shape	*Diagram*	*Branches*	*Bonds*	*Subunit*	*Source*	
				α- glucose		Amylose
		yes per 10 glucose				Amylopec-tin
			1-4 and 1-6			Glycogen
						Cellulose
						Chitin

Facts

- The purest form of cellulose found in nature is cotton. It contains 95% of cellulose.
- Starch is the most important source of carbohydrates in the human diet. Greater percentage of starch is found in potato 15%, wheat 55%, corn 65%, and rice 75%.

Research time

Write a report about biological role of each polysaccharide.

Literacy

1. Which polysaccharide is the most abundant in nature?
2. Describe how the structure of cellulose plays an important role in plants as a component of cell walls?
3. Glycogen is composed of short highly branched chains with much accessible glucose. Why is this important for its function.

Terminology

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- accessible - қолжетімді / доступный;
- branched - тармақталған / разветвленный;
- cellulose - жасуық / целлюлоза;
- chitin - хитин;
- fibrous - талшықты / волокнистый;
- glycogen - гликоген;
- helical - шиыршықтанған / спиральный;
- linear - сызықтық / линейный;
- plane - жазықтық / плоскость;
- starch - крахмал.

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1.4 Reducing and Nonreducing sugars

You will:

- identify reducing and nonreducing sugars.

Stimulating question

How do doctors find carbohydrates during urine analysis of patients with diabetes?

Key terms

- Reducing sugar - sugar that serves as a reducing agent due to its aldehyde or ketone group in its molecular structure;
- Nonreducing sugar - sugar that cannot donate electrons to other molecules and therefore cannot act as a reducing agent;
- Reducing agent - reduces other substances and loses electrons;
- Oxidizing agent - oxidises other substances and gains electrons.

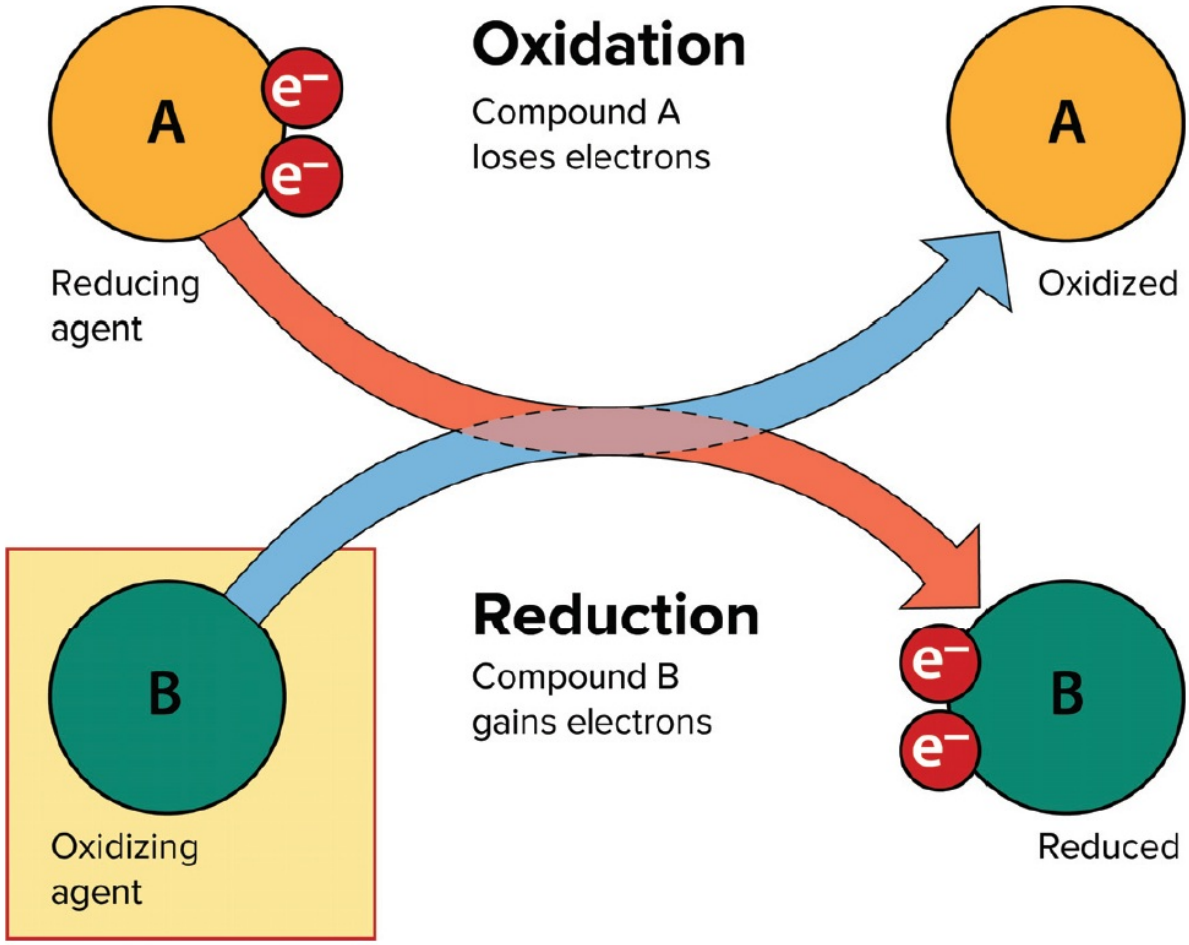
Text

Carbohydrates also classified as reducing and nonreducing sugars. Sugar that contains a free aldehyde or ketone group is called reducing sugar.

In reducing sugar, free aldehyde or ketone group acts as a reducing agent in a redox reaction. Redox (oxidation-reduction) reaction is a type of chemical reaction that involves the transfer of electrons between two molecules. Reducing agent donates electrons to another molecule. The molecule which receives electrons is called an oxidising agent.

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Redox reactions

Nonreducing sugars do not contain free aldehyde or ketone group, and therefore they cannot donate electrons.

Benedict's test is used to identify whether carbohydrate is reducing or nonreducing sugar. It is a chemical reagent which is a complex mixture of sodium carbonate, sodium citrate and copper (II) sulphate pentahydrate. When reducing sugars are mixed with Benedict's solution and heated, heating causes Benedict's solution to change colour because copper (II) ions are reduced to copper (I) ions.

Labwork

Pre-lab questions:

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1. What is the monomer of starch?
2. What is the function of amylase?
3. Does glucose molecule have aldehyde or ketone group?

Methods and Materials:

table sugar, starch, water, saliva, four test tubes, water bath, Benedict's solution, droppers, tweezers.

Safety precautions:

1. Be careful with boiling water.
2. Do not taste the materials.

Procedures:

1. Mix starch with water to make a starch solution and place 1 ml of it into the first and second test tubes.
2. Add saliva into the first test tube. Wait for 5 minutes.
3. Mix table sugar with water to make sucrose solution.
4. Place 1 ml of sucrose solution into the third test tube and 1ml of water into the fourth test tube.
5. Pour 10 drops of Benedict's solution to every sample.
6. Place all test tubes into the water bath for 10 minutes.
7. Observe color change in each sample and fill the table.

Results:

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Test samples	*Carbohy-drates*	*Color*	*Reducing or Nonreducing sugars*	*Chemical structure*
1st test tube				
2nd test tube				
3rd test tube				
4th test tube				

Post-lab questions:

1. Why the first test and second test tubes have different colors? Explain your answer.
2. Benedict's test can be used to test for the presence of glucose in urine. Why it is needed?

Facts

All monosaccharides are reducing sugars.

Research time

Find out three reducing and nonreducing sugars. Draw their chemical structure and prove if they are reducing or nonreducing sugars.

Terminology

- interpretation - түсінік беру / толкование;
- nonreducing sugar - тотықсызданбайтын қант / нередуцирующийся сахар;
- oxidation - тотығу / окисление;
- reduction - тотықсыздану / восстановление;
- reducing sugar - тотықсызданатын қант / редуцирующийся сахар;
- to donate - беру / давать;

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- to oxidize - тотықтыру / окислять;
- to reduce - тотықсыздандыру / восстанавливать.

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1.5 Lipids

You will:

- describe the chemical structure and function of lipids.

Stimulating question

How can camels survive so long without water?

Key terms

- Lipids - biomolecules that are soluble in organic solvents and insoluble in water;
- Triglyceride - an energy-rich compound made up of a single molecule of glycerol and three molecules of fatty acid

Text

Lipids are organic compounds that consist of carbon, hydrogen and oxygen atoms. Some lipids may contain phosphorus and nitrogen. Unlike carbohydrates, lipid molecules have much lower proportion of oxygen to hydrogen.

Lipids are nonpolar molecules, they cannot be dissolved in water. However, they are soluble in nonpolar solvents, such as alcohol. They are very diverse, and they have variety of structures and functions. The most well-known lipids are fats and oils which referred to triglycerides.

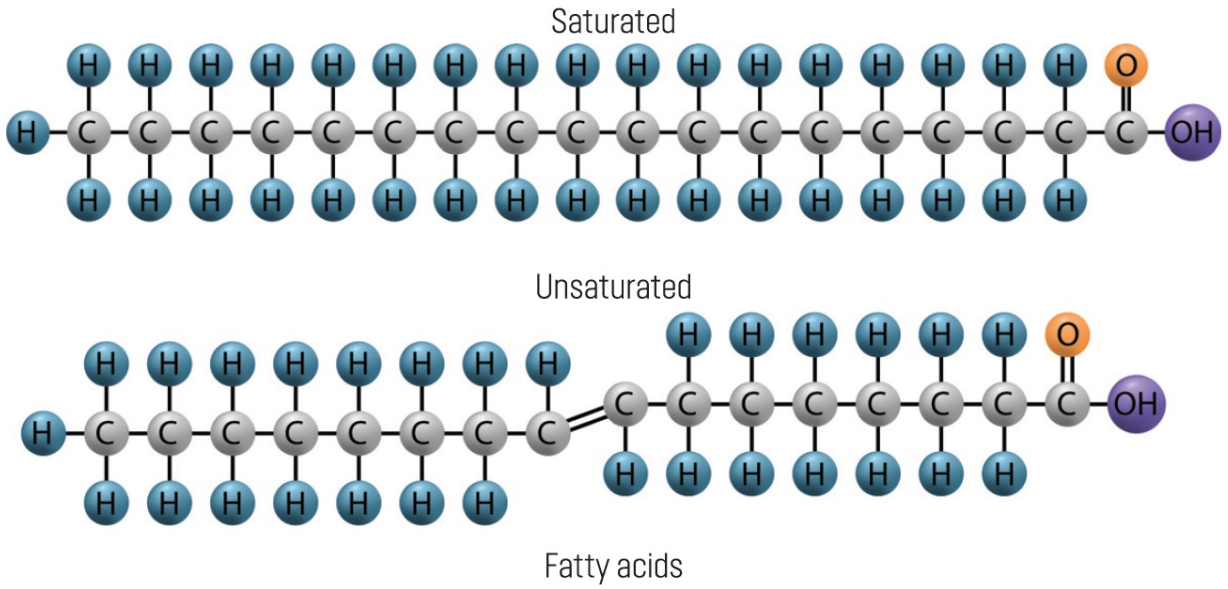
Triglycerides

Triglycerides contain glycerol and three fatty acids. Glycerol is a three carbon-containing molecule. It has three polar hydroxyl (-OH) groups, which makes it soluble in water.

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Glycerol always has constant structure, while fatty acids have different forms. According to this, triglycerides differ in structure and function. A fatty acid has a long chain of carbon atoms bonded only to hydrogen and carboxyl group at one end. They are classified as saturated and unsaturated fatty acids. Saturated fatty acids have single bonds (C - C) in the hydrocarbon chain, and they are saturated with hydrogen. Unsaturated fatty acids have fewer hydrogen atoms because they have one (monounsaturated) or more (polyunsaturated) double bonds (C = C) in the hydrocarbon chain.



To form triglyceride, each of hydroxyl (-OH) group in glycerol reacts with carboxyl (-COOH) group of fatty acids. This reaction removes three molecules of water and creates ester bonds between glycerol and fatty acids.

Fats consist of saturated fatty acids. They are solid at room temperature and produced in animals.

Oils generally have a higher proportion of unsaturated fatty acids. They are liquid at room temperature and mostly found in plants.

Double bonds in fatty acids create a bend in a hydrocarbon chain. These bends prevent close packing of hydrocarbon chains and explain fluidity of oil.

Lipids are used as high energy stores because they are insoluble in water and stable. They can be

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easily used to release energy.

Fats form insulating layer against heat loss. They conduct heat energy very slowly and help organisms to keep warm in cold climates. In addition, fat under skin protects the internal organs against physical damage and provide buoyancy.

Fat is often deposited near the delicate organs like liver and heart to give extra protection.

Activity

1. Make a Venn diagram of fats and oils. Give examples of food products rich in fat.
2. Draw a saturated and an unsaturated fatty acids with six carbons. Count hydrogen numbers in each of them.
3. Draw a simplified diagram of triglyceride. Label glycerol, fatty acids and ester bonds.
4. As the number of double bonds in fatty acids increase, the melting point decreases. If you put a product with high level of unsaturated fats and a product with high level of saturated fats in a refrigerator, how would they differ in consistency? Explain your answer.

Facts

- Lipids release many water molecules when they are metabolized. The camel’s hump contains fat which produces useful amount of water.
- There are two essential fatty acids, linoleic and linolenic acids, that humans cannot synthesise. They must acquire these fatty acids from food like fish, broccoli, nuts, olive oil, and grain.

Literacy

1. What is the biological importance of triglycerides?
2. Which properties of lipids help them play their biological roles? Explain your answer.
3. How fat layer provides buoyancy?

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Research time

1g of lipid can yield over twice as much energy as 1g of carbohydrate and protein. Do a research and explain this statement.

Terminology

- buoyancy - қалқымалылық / плавучесть;
- conduct - өткізу / проводить;
- consistency - консистенция;
- fatty acid - май қышқылы / жирная кислота;
- glycerol - глицерин;
- saturated - қаныққан / насыщенный;
- to create - жасау / создавать;
- to dissolve - еру / растворяться;
- to insulate - оқшаулау / изолировать;
- triglyceride - триглицерид;
- unsaturated - қанықпаған / ненасыщенный.

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1.6 Classification of Lipid

You will:

- describe the chemical structure and function of lipids.

Stimulating question

Which substances do bees use to build honeycombs? How bees build their honeycombs?

Key terms

- Phospholipid - a lipid containing a phosphate group in its molecule;
- Steroid - a type of lipid characterised by carbon skeleton consisting of four fused rings with various chemical groups attached;
- Wax - a lipid made up of a chain of alkanes or esters from alcohols and fatty acids

Text

Apart from fats and oils, there are other types of lipids which play an important role in living organisms. They include phospholipids, waxes, and steroids.

Phospholipids

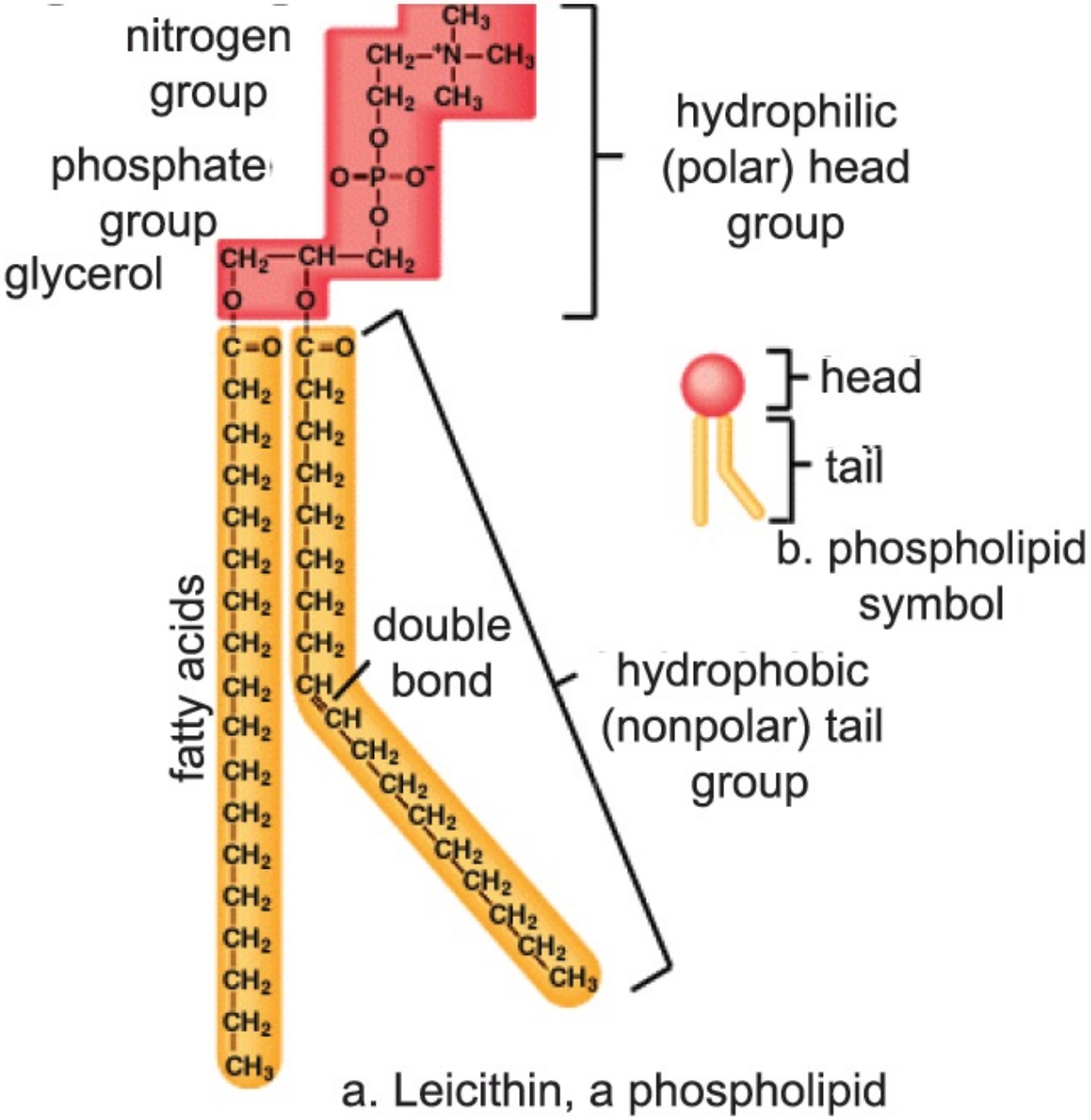
Phospholipids include phosphorus (P) in addition to C, H, and O. They are formed by condensation of glycerol with two fatty acid molecules and phosphoric acid molecule. Phosphoric acid reacts with one of the three hydroxyl groups (-OH) of glycerol while other two hydroxyl groups (-OH) react with fatty acid chains. A phospholipid has two distinct regions: polar head and nonpolar tails. The “phosphoric” part of the molecule has polar hydroxyl groups (-OH) groups which make it hydrophilic (water-loving). Hydrophilic head is attracted to water molecules. Another part of the molecule including two fatty acids is

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hydrophobic (water-hating). Hydrophobic tails repel water molecules.

In aqueous solution, phospholipids form spherical structures called micelles. The hydrophobic tails turn inwards and become protected from water by hydrophilic heads. These properties determine the structure and function of the cell membrane. Cell membrane consists of mainly phospholipids bilayer. It forms a boundary between the outside and inside environment of the cell.



Waxes

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Wax is composed of a single long-chain fatty acid linked to a long-chain alcohol other than glycerol. They are very hydrophobic because of their long nonpolar chains, and animals cannot digest them.

Waxes form waterproof coverings in living organisms. These covers stop water from getting in, and for terrestrial plants and animals, it prevents water loss. For example, feathers of birds are coated with wax to repel the water. Also, insects have a waxy cuticle to prevent water loss.

Steroids

Steroids consist of four interlinked rings of carbon atoms with various side chains. Cholesterol is a well-known steroid. It is produced in the liver as well as being supplied in the diet. It is a vital component of the cell membrane which helps to keep it fluid. All other steroids and vitamin D are synthesised from cholesterol. Some hormones are made up of steroids such as aldosterone, progesterone, and testosterone.

Activity

Identify the differences between types of lipids. Fill the table.

Lipids	*Components*	*Examples*	*Functions*
Triglycerides			
Phospholipids			
Waxes			
Steroids			

Facts

- Bees have ceruminous glands which secrete wax for building their honeycombs.
- Anabolic steroids are substances similar to testosterone which promote muscle growth. The use of these drugs forbidden in international sports.

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They cause many health diseases such as liver damage and heart diseases.

Literacy

1. What is the biological importance of triglycerides?
2. Which properties of lipids help them play their biological roles? Explain your answer.
3. How fat layer provides buoyancy?

Research time

Cholesterol is often related to heart diseases. Do a research and identify its connection. Use words below and write a report.

lipoprotein	LDL	HDL
blood flow	atherosclerosis	
stroke	heart attack	diet

Terminology

- ceruminous glands - церуминозды бездері / церуминозные железы
- cuticle - кутикула;
- distinct - айқын / отчетливый;
- forbidden - тыйым салынған / запрещенный;
- honeycomb - кәрез / соты;
- interlinked - өзара байланысқан / взаимосвязанный;
- to consider - қарастыру / рассматривать;
- to determine - анықтау / определить;
- to promote - мүмкіндік туғызу / способствовать;
- to repel - итеру / отталкивать;
- waterproof - су өтпейтін / водонепроницаемый;
- wax - балауыз / воск.

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1.7 Protein

You will:

- classify proteins according to their composition, structure and functions.

Stimulating question

What makes spider silk strong and stretchy?

Key terms

- Amino acid - a simple organic compound containing both a carboxyl (—COOH) and an amino (—NH2) group;
- Protein - large biomolecules, or macromolecules, consisting of one or more long chains of amino acid residues.

Text

Protein is the most important polymer, which consists of amino acids. All amino acids contain carbon, hydrogen, oxygen and nitrogen atoms while some may have sulphur atoms too.

Amino acids have the basic structure which includes a central carbon atom, bonded to an amino group (−NH2), a carboxyl group (−COOH), and a hydrogen atom. Also, there is another atom or group of atoms linked to central carbon known as radical group (R-group). R-groups differ in amino acids and determine each amino acid.

Amino acids link together by condensation reaction. When two amino acids condense, the amino group of one amino acid links with the carboxyl group of another amino acid forming a dipeptide. The bond which links amino acids is called a peptide bond. Continuous condensation reactions between amino acids create long chains of amino acids called polypeptides. Proteins consist of one or

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more polypeptides.

Proteins are classified according to their composition, structure and function.

Based on chemical composition, proteins divided into two groups: simple and conjugated proteins. Simple proteins contain only amino acids. Examples are plasma albumin, collagen and keratin. Conjugated proteins include a non-protein portion called as a prosthetic group. The prosthetic group is a compound that helps proteins in different ways. According to the prosthetic group, conjugated proteins subdivided into:

- Glycoproteins contain carbohydrates. Ex: blood plasma proteins (except albumin), immunoglobulin, and antibodies.
- Lipoproteins contain lipids.They are found in the bloodstream and as a component of the plasma membrane.
- Phosphoproteins contain a phosphate group. Ex: milk casein and egg white ovalbumin.
- Chromoproteins contain pigments. Ex: hemoglobin which contains heme, rhodopsin which contains retinal.

Proteins have four levels of structure.

Primary structure

Primary structure of a protein is the sequence of amino acids in the chain. This sequence determines proteins shape, and shape determines its function. The only bonds involved in this structure are peptide bonds.

Secondary structure

The secondary structure of a protein involves the twisting or folding of polypeptide chains caused by hydrogen bonds. The backbone of polypeptides carries partially positive and negative charges which form hydrogen bonds.

Common types of secondary structure are alpha-helices and beta-pleated sheets.

In the alpha-helix, the CO group of one amino acid is hydrogen bonded to NH group of amino acid further along the chain. These bondings twist the chain into the spiral shape and help to maintain the shape of the protein.

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Beta-pleated sheets formed in anti-parallel polypeptide chains. CO groups of one chain are hydrogen bonded to NH group of parallel chains folding a flat structure of the protein. This shape gives strength to the protein.

Tertiary structure

The tertiary structure of a protein involves the folding and coiling of polypeptide chains into three-dimensional shape. It is formed by various bonds between R-groups of amino acids. Different shapes can be created by different R-groups and by three types of bonds: hydrogen, ionic, and disulphide bonds.

Hydrogen bonds are weakest bonds formed between H and O atoms.

Ionic bonds occur between any charged groups that are not joined by the peptide bond. These bonds are stronger than hydrogen bonds, however weaker than disulphide bonds.

Disulphide bonds form between sulfur atoms found in R-groups.

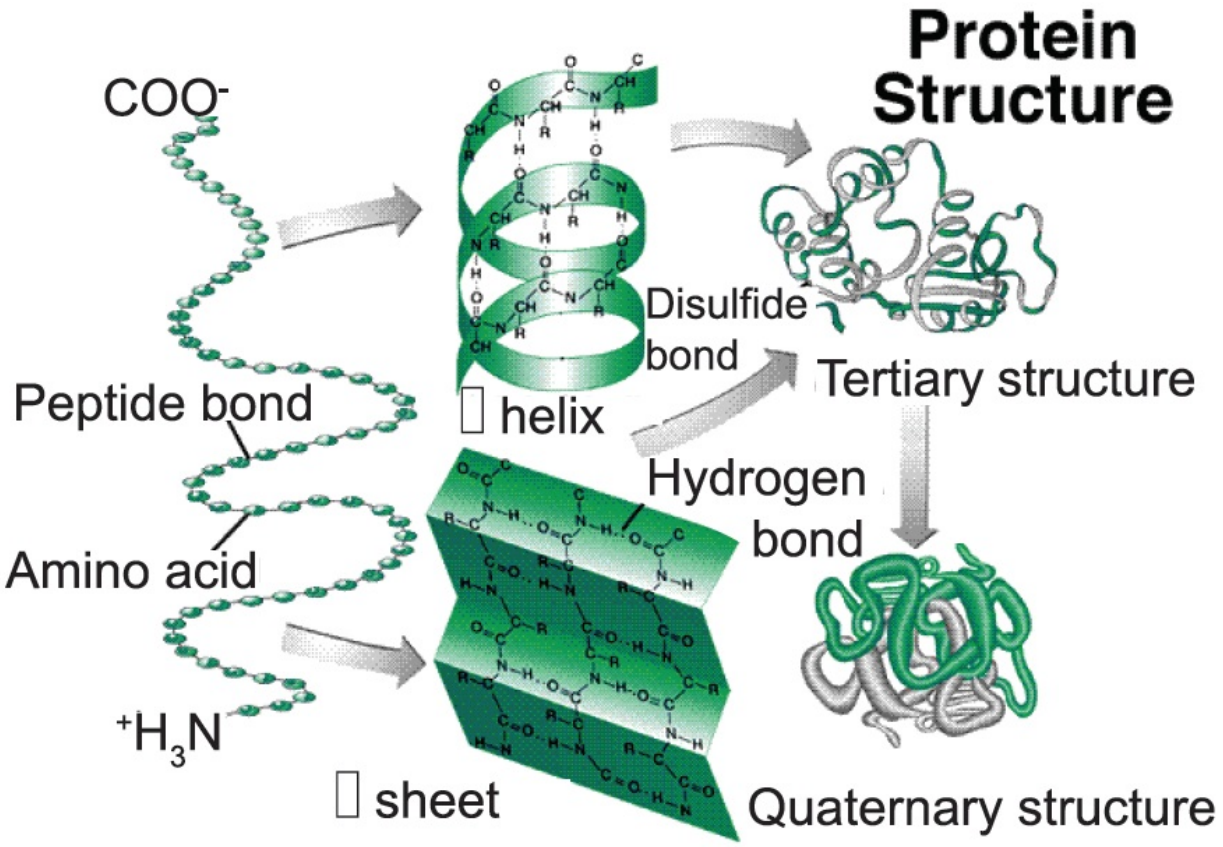
Also, the shape of protein is affected by hydrophobic R-groups which repel the water.

Quaternary structure

The quaternary structure of a protein involves two or more polypeptide chains linked together. Each polypeptide chain is called a subunit of the protein.

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Activity

1. There are 20 different amino acids, which make up proteins. Polypeptide chains may be formed with any number and any different order of amino acids. How many different polypeptides would be formed if the chain contains hundred of amino acids?
2. Compare each levels of protein structure and fill the table.

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Levels	*Defini-tion*	*Simple diagram*	*Bonds involved*	*Impor-tance*	*Examples*
Primary structure					
Secondary structure					
Tertiary structure					
Quaternary structure					

Facts

Spider dragline silk has regions of alpha-helix and beta-pleated sheet that make it both elastic and strong.

Literacy

1. What is the reaction by which proteins are formed from amino acids?
2. Explain the basis for the great diversity of protein?

Research time

A small change in primary structure of protein crucially affects its shape and function. Do a research about sickle cell anemia. Explain its cause and write a report about this disease.

Terminology

- alpha-helix - альфа-шиыршық / альфа-спираль;
- amino group - амин тобы / аминная группа;
- beta-pleated sheet - бета-қатпарлы қабат / бета складчатая форма;
- carboxyl group - карбоксил тобы / карбоксильная группа;

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- coiling - бүктелу / свертывание;
- conjugated - күрделі / сложный;
- folding - бүгіліс / сгиб;
- prosthetic group - простетикалық топ / простетическая группа;
- radical group - радикал тобы / радикальная группа;
- to condense - топталу / конденсироваться;
- to subdivide - бөліну / подразделять;
- twist - бұрылыс / поворот;

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1.8 Functions of Protein

You will:

- classify proteins according to their composition, structure and functions.

Stimulating question

Why does skin wrinkle with age?

Key terms

- Globular proteins - proteins coiled into compact shapes with hydrophilic outer surface that make them water soluble;
- Fibrous proteins - proteins composed of long linear polypeptide chains that are bundled together to form strand or sheets.

Text

Proteins perform different vital functions in the organisms. Their functions are directly related to their shapes. Proteins can be classified depending on their final three-dimensional structure: globular and fibrous proteins.

Globular proteins

In globular proteins, polypeptide chains are folded and coiled tightly to form a spherical shape. Each of them has its specific shape and length of polypeptide chains. Globular proteins are unstable molecules. They are soluble in water and can be easily changed chemically. Their main functions are metabolic. Classes of globular proteins are:

- Enzymes bring reactants together and act as a catalyst. They speed up chemical reactions in the cell.
- Antibodies are protective proteins. They destroy different antigens

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protecting the body against disease;
- Transport proteins, such as hemoglobin, carry oxygen in the blood. Channel proteins in plasma membrane allow substances enter and exit the cells;
- Storage proteins, such as myoglobin, store oxygen in muscle;
- Receptor proteins in plasma membranes and inside the cell recieve signals from complementary molecules and trigger particular reactions;
- Hormones are regulatory proteins. Insulin regulates the concentration of glucose in the blood;

Fibrous proteins

In fibrous proteins, polypeptide chains are parallel to each other. They are linked together to form long fibres or sheets. Fibrous proteins are insoluble in water and very stable molecules. These proteins provide structure and support for cells. Classes of fibrous proteins are:

- Structural proteins give support. Keratin is found in hair, skin, nails, feathers, hooves and horns. Collagen gives flexible strength to tendons, blood vessel walls and skin. Elastin provides strength and elasticity to ligaments;
- Contractile proteins are involved in the movement. Actin and myosin cause the contraction of muscles.

Activity

a. Compare globular and fibrous proteins. Write four characteristics to each type of protein.

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Globular proteins	Fibrous proteins

b. Compete with classmates by finding as much as possible proteins found in human body. Name and label them in the diagram and explain their functions.

Facts

- Whales' muscles have higher concentration of myoglobin which allows them to remain underwater for long periods of time.
- After the age of 20, a body produces less amount of collagen and elastin in the skin year by year. As a result, the skin becomes thinner and more fragile with age.

Literacy

1. Which levels of organisation of proteins refer to globular proteins?
2. What is the function of triple helix structure in collagen?

Research time

Deficiency of vitamin C cause scurvy. This disease include the loosening of teeth and internal bleeding. Explore the connection between vitamin C and collagen. Write a report.

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Terminology

- fibrous - талшықты / волокнистый;
- fragile - нәзік / хрупкое;
- globular - глобулярлы / шаровидный;
- hoof - тұяқ / копыто;
- horn - мүйіз / рог;
- ligament - байлам / связки;
- loosen - разрыхлять / жұмсарту;
- scurvy - цинга;
- tendon - сіңір / сухожилия;
- to destroy - жою / уничтожать;
- to remain - қалу / оставаться;
- unstable - тұрақсыз / неустойчивый;
- wrinkle - морщина / әжім.

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1.9 Protein Denaturation

You will:

- investigate the effect of different conditions on the structure of proteins.

Stimulating question

Why do human cook many kinds of food before eat them?

Key terms

- Denaturation - process of altering the shape of a protein without breaking the peptide bonds that form primary structure.

Text

Proteins consist of amino acids. Radical groups (R-groups) of various amino acids interact with each other and form hydrogen, ionic or disulfide bonds. These bonds hold the natural shape of proteins. However, different conditions can easily break these bonds.

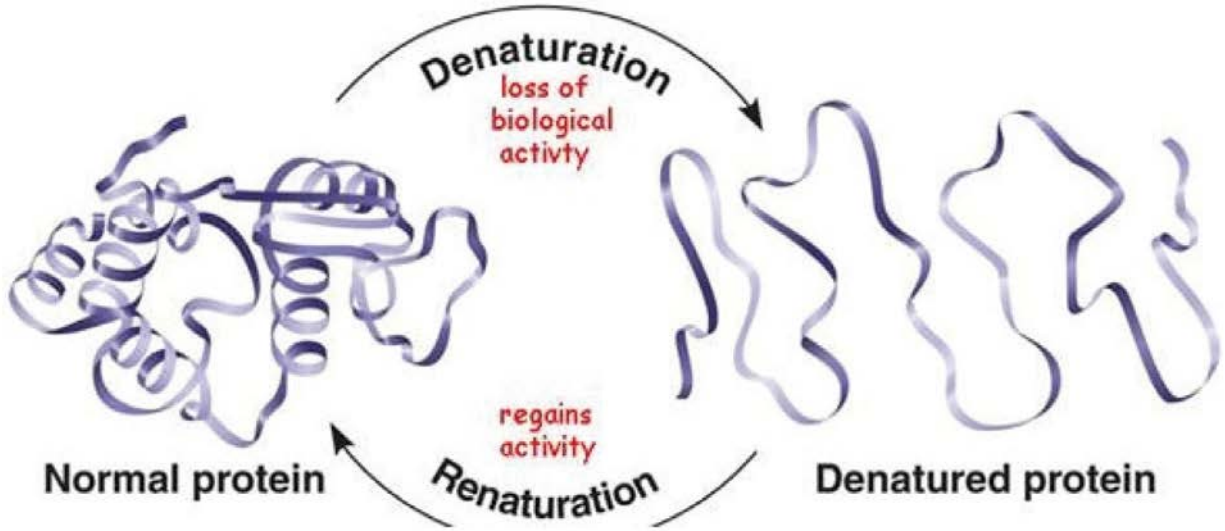
A rise in temperature causes the molecules to vibrate rapidly. It disrupts hydrogen bonds and hydrophobic interactions and changes protein’s overall shape.

Also, pH changes in solution affect hydrogen bonds and ionic bonds. Changes in concentration of the positively charged hydrogen ions alter the forces of attraction holding the protein molecule shape.

When proteins lose their folded structure and cannot function, they denature. Denaturation of proteins involves the disruption of quaternary structure, tertiary structure and secondary structure. The primary structure remains constant after denaturation process because peptide bonds are not affected.

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Denaturation and renaturation

Albumin is a protein found in egg white. Cooking or whipping egg whites denatures the globular proteins changing egg white from clear to white forming insoluble protein.

Sometimes denaturated proteins can reconstruct into the original form. This process is called renaturation. Renaturation is rare process. Some proteins may be renatured if the denaturing agent or molecule is removed.

Labwork

Pre-lab questions:

1. Which types of bonds are weak in protein?
2. What is pH?
3. Why it is important to keep constant body temperature?

Methods and Materials:

3 egg whites, water, lemon juice, NaHCO3 (baking powder), 4 test tubes, water bath, stirring rod.

Safety precautions:

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1. Be careful with boiling water.
2. Do not taste the materials.

Procedures:

1. Label the test tubes from 1 to 4.
2. Separate egg white and egg yolk. Discard egg yolk.
3. Place egg white into each test tubes until half filled.
4. Place the test tube 2 in the boiling water till it becomes white.
5. Add one teaspoon of lemon juice to the test tube 3 and stir.
6. Add NaHCO3 to the test tube 4 and stir.
7. Compare test tubes with test tube 1.
8. Record observations to the table below.

Results:

Test tubes	*Added*	*Observation*
1		
2		
3		
4		

Post-lab questions:

1. Which method has the greatest denaturing effect on egg albumin? Explain your answer.
2. Why heat and alcohol are used to disinfect medical equipment?
3. Why peptides bonds are not affected by denaturation?

Facts

- Some enzymes such as trypsin cannot access to peptide bonds due to natural shapes of proteins. Denaturation of proteins makes peptide bonds accessible to split them and helps in digestion..

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- The layer that forms on curdled milk is another common example of denatured protein..

Research time

Do research about renaturation. Find some examples of renaturation of proteins.

Terminology

- baking powder - наубайхана ұнтағы / пекарский порошок;
- denaturation - денатурация;
- influence - әсер ету / влияние;
- rapidly - тез / быстро;
- to alter - өзгерту / изменять;
- to discard - тастау / отбрасывать;
- to disrupt - бұзу / разрушать;
- to interact - өзара әрекеттесу / взаимодействовать;
- to lose - жоғалту / терять;
- to stir - араластыру / размешивать.

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1.10 Protein Determination

You will:

- determine the protein content in biological objects.

Stimulating question

How forensics investigators determine the fingerprints of criminals?

Key terms

- Aromatic rings - are hydrocarbons that contain benzene;
- Ninhydrin reagent - chemical that reacts with free amino groups of amino acids.

Text

Various protein determination methods are used in different fields. Determination of protein in urine or in blood plasma help to identify the disease. Food analysts do protein determination methods to know total concentration, type, molecular structure and functional properties of the proteins in foods.

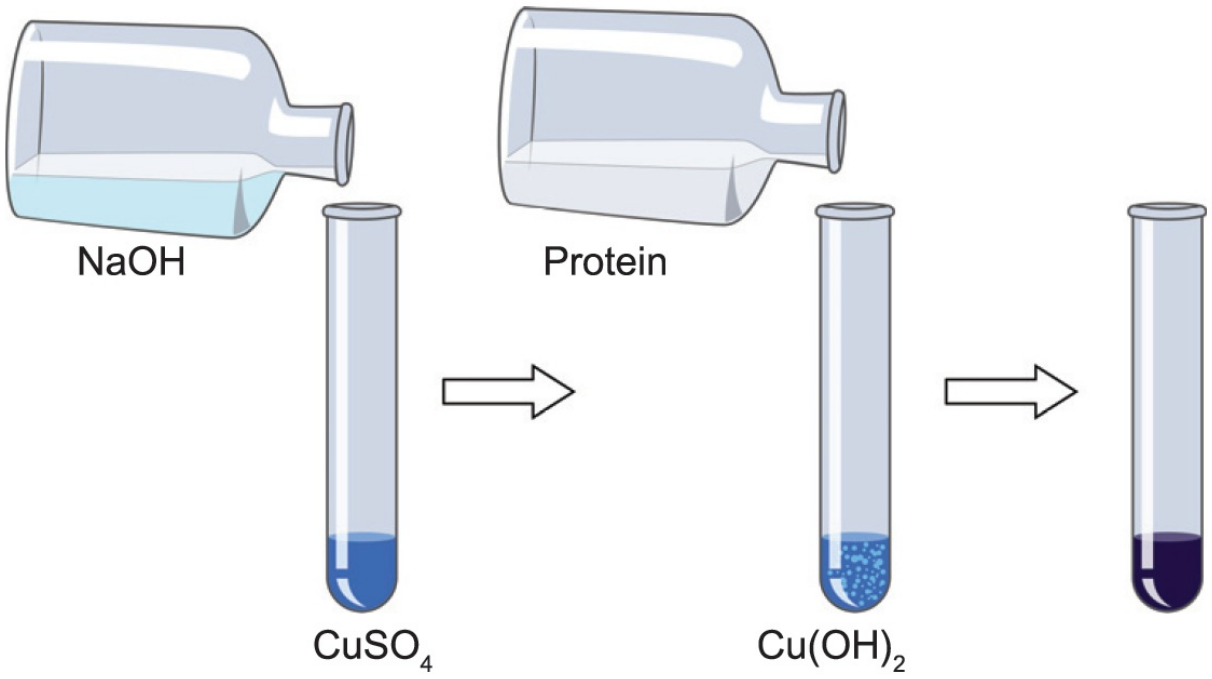
The presence of proteins in biological objects or solutions can be determined by color reactions, the course of which is due to the presence of specific groups and peptide bonds in the protein. There are some common methods which are used to determine proteins.

Biuret Test

Compounds that contain two or more peptide bonds will react with Cu^{2+} in a basic solution to form a violet-pink complex. The original Cu^{2+} solution is blue, so if the solution remains blue, the compound being tested could be an amino acid or a dipeptide or neither.

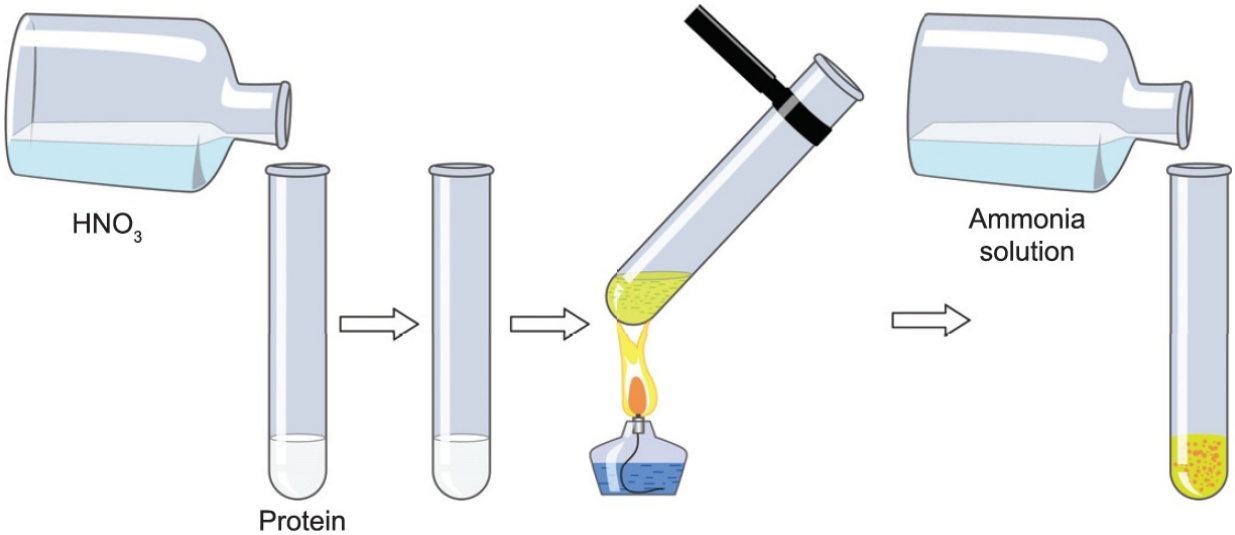
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Xanthoproteic Test

The aromatic rings on tyrosine and tryptophan react with nitric acid. In this reaction, the aromatic rings become nitrated. When nitric acid is added to a sample and the mixture is heated, a yellow solution will result if the sample contains tyrosine or tryptophan amino acids. When this yellow solution is treated with a strong base (such as NaOH), it turns orange. Since most proteins contain one or both of these amino acids, most proteins will show a positive reaction in this test.



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Ninhydrin Test

Free amino groups will react with the ninhydrin reagent to yield a purple solution. Almost all amino acids contain a free amino group (except proline). Some proteins also give a positive test with ninhydrin.

Facts

- Biuret solution is blue liquids which used for protein assay. The copper atom of the biuret solution reacts with the peptide bonds to cause the colour change. It changes to purple when proteins are present and to pink in the presence of short chains of polypeptides.
- A ninhydrin solution is commonly used by forensic investigators in the analysis of latent fingerprints on porous surfaces such as paper. It was first used in 1959 in forensics.

Research time

If you had completely hydrolyzed the egg albumin before doing the Biuret test on that sample, what results would you expect for the Biuret test? Explain why. Include what color you would expect to see, whether it is a positive or negative test, and what that means.

Activity

Pre-lab questions:

1. Which protein found in milk?
2. Which protein found in egg?
3. Can denaturation split peptide bonds?

Methods and Materials:

biuret reagent, water, milk, egg white, apple, cracker, 5 test tubes, dropper.

Safety precautions:

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1. Wear gloves.

Procedures:

1. Label test tubes from 1 to 5.
2. Separate egg white and egg yolk. Discard egg yolk.
3. Place water in 1 test tube.
4. Place egg white into 2 test tubes.
5. Place milk in 3 test tube.
6. Crush apple piece and cracker.
7. Place crushed cracker into test tube 4 and add some water.
8. Place crushed apple piece into test tube 5 and add some water.
9. Add about 2 ml of biuret reagent to each test tube. Wait for 10 minutes.
10. Observe results and fill the table below.

Resluts:

Test tubes	*Added*	*Colour*	*Protein pres-ence*	*Protein amount*
1				
2				
3				
4				
5				

Post-lab questions:

1. Which test tube has short chains of polypeptides?
2. Which test tube has a high amount of protein?
3. Can biuret reagent stain your skin and fingernails? Explain your answer.

Terminology

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- aromatic rings - ароматтық сақиналар / ароматические кольца;
- assay - талдау / анализ;
- determination - анықтау / определение;
- forensics - сот-медициналық сараптама / судебно-медицинская экспертиза;
- latent fingerprints - жасырын саусақ іздері / скрытые отпечатки пальцев;
- nitric acid - азот қышқылы / азотная кислота;
- to determine - анықтау / определять.

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1.11 DNA

You will:

- establish a connection between the structure of deoxyribonucleic acid and its function.

Stimulating question

How organisms differ if their DNA include same components?

Key terms

- DNA - a molecule which stores the genetic information of an organism and transmits that information from one generation to another;
- Nucleotide - the basic building block of nucleic acid polymers; an organic compound made up of a nitrogenous base, a sugar, and a phosphate group.

Text

Deoxyribonucleic acid (DNA) is a type of nucleic acid which found in every cell. It stores genetic information and transfers it from one generation to another.

DNA consists of nucleotides. Each nucleotide contains three components: sugar, a phosphate group, and a nitrogen-containing base. Sugar, deoxyribose, contains five carbon atoms. The phosphate group comprises phosphoric acid (H3PO4).

The third component includes four different bases: adenine (A), guanine (G), cytosine (C), thymine (T). Adenine and guanine have a double ring structure, and they are purines. Thymine and cytosine have a single ring structure, and they are pyrimidines.

Three components link by condensation reactions to form a nucleotide. A pentose sugar attaches to phosphoric acid at 5’, and to one of the bases at 1’.

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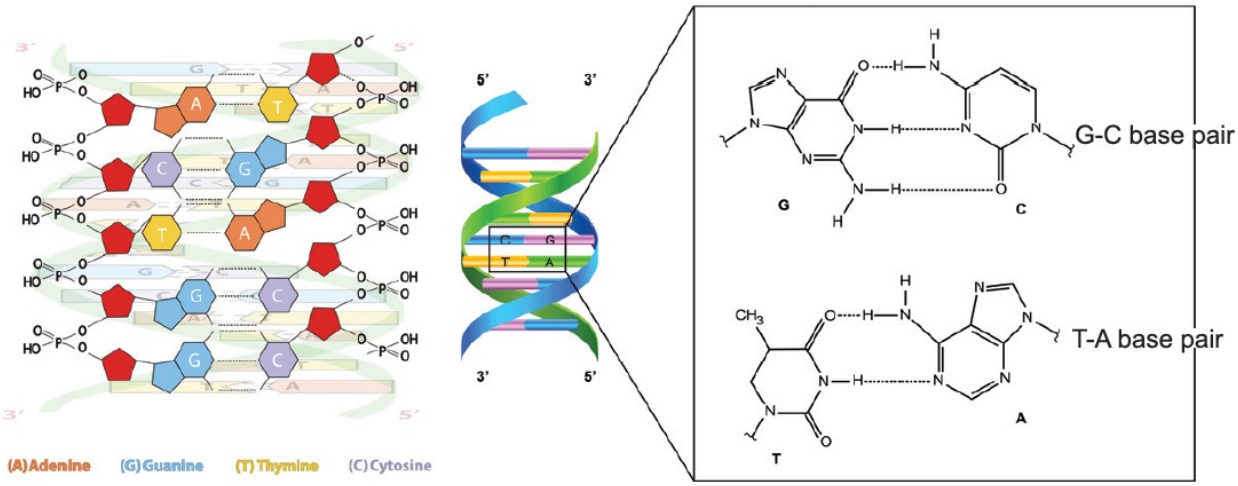
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Primes (‘) are used to number the carbons in pentose sugar. The linkage of nucleotides occurs when the 3’-OH of pentose sugar in one nucleotide joins to the 5’- phosphate of a second nucleotide forming a phosphodiester bond. Phosphodiester bonds form polynucleotide chains. The sequence of nucleotides in polynucleotide chain is the primary structure of DNA. It determines the genetic information of the organism.

The secondary structure of DNA is a double helix structure. DNA consists of two polynucleotide chains that twisted around each other. It has a ladder-like structure where sugar and phosphate group form its backbone while bases form its rungs. Two strands are antiparallel to each other. One strand starts from 5’ and ends with 3’, while the other strand starts from 3’ and ends with 5’. Nitrogenous bases are linked to each other by hydrogen bonds, and they can only bond with a specific base partner forming a complementary base pairing. Adenine pairs with thymine using two hydrogen bonds. Guanine pairs with cytosine using three hydrogen bonds.

DNA stores genetic information in the sequence of nucleotides. Nucleotides are complementory paired to each other. It helps to transfer genetic information to the next generations.

This spiral arrangement and hydrogen bonding between bases makes DNA stable. It helps to protect nucleotide sequence from being affected and changed by the environment.



Hydrogen bonding between base pairs

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Facts

- The double helix structure of DNA discovered in 1953 by James Watson and Francis Crick. They were awarded a Nobel prize in 1962.

Research time

The greatest scientific discovery of 20th century is double helix structure of DNA. Apart from Watson and Crick, other scientists made significant contributions to this invention. Explore the history of this discovery and fill the table.

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Scien-tist	*Contri-bution*	*Year*
Erwin Char-gaff		
Rosa-lind Frank-lin		
Mau-rice Wilkins		

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Literacy

1. To which group of carbohydrates does deoxyribose belong?
2. Hydrogen bonds are weak bonds. How can two strands of DNA not split?
3. Look at two DNA sequences below. Determine which DNA is more stable. Explain your answer.

(a) 5’AATTTAAGGGCCCAAATTTT3’ (b) 5’CACGTAATGCATTCGCCAGT3’

3’TTAAATTCCCGGGTTTAAAA5’
3’GTGCATTACGTAAGCGGTCA5’

Terminology

- arrangement - орналасуы / расположение;
- contribution - үлес / вклад;
- double helix - қос спираль / двойная спираль;
- invention - өнертабыс / изобретение;
- phosphodiester bond - фосфодиэфирлік байланыс / фосфодиэфирная связь;
- purines - пурин;
- pyrimidines - пиримидин;
- to comprise - қамту / включать;
- to discover - жаңалық ашу / делать открытие;
- to explore - зерттеу / исследовать.

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1.12 DNA Replication

You will:

- describe the process of DNA replication based on Chargaff’s rule.

Stimulating question

Why are daughter cells identical to the parent cell?

Key terms

- DNA replication - process by which DNA makes a copy of itself before cell division.

Text

DNA transfers genetic information from one generation to another. It has to copy itself to allow offspring inherit genes from parents. The process by which DNA copies itself is called replication. DNA replication occurs by semi-conservative model. It means each new generation inherit DNA that contains one old strand and one new strand. The semi-conservative model of replication of DNA was proved by two scientists Matthew Meselson and Franklin Stahl.

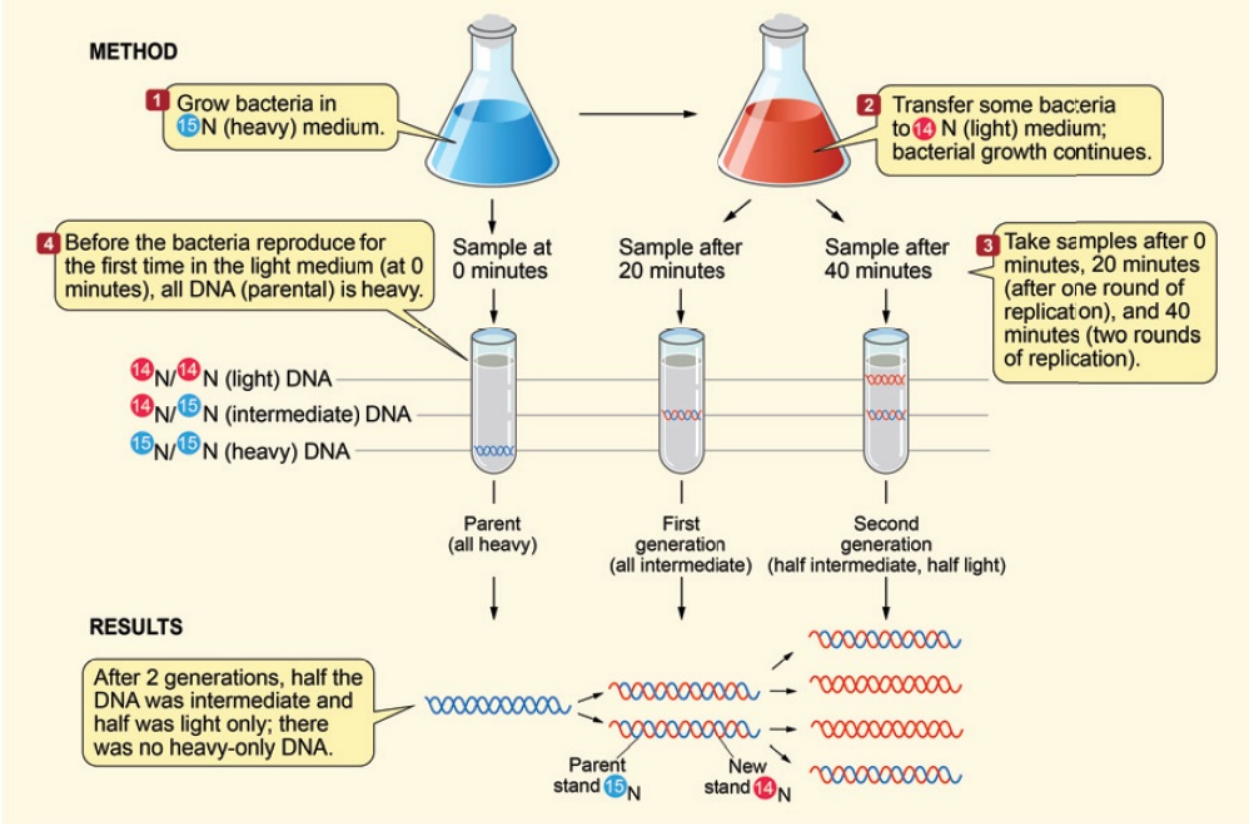
Meselson-Stahl experiment

Meselson and Stahl made an experiment involving E.coli. Bacteria were grown on a medium in which normal nitrogen (14N) was replaced with the heavy isotope (15N). Isotope is any of two or more forms of a chemical element, having the same atomic number, but having different atomic weights. The molecules containing 15N are heavier than those containing 14N. After many generations passed, the medium containing 15N was replaced on a medium containing 14N. Then samples of bacteria were taken from the culture after one and after two generations. Ultracentrifugation was used to separate the DNA

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molecules according to the ratio of 15N and 14N.



Meselson-Stahl experiment

In medium containing 15N, DNA was heavy. After replacing with normal 14N, the first generation indicated DNA was intermediate, between 15N and 14N. The second generation detected two types of DNA. One type had only 14N while another was the same as in the first generation, half 15N and a half 14N. Results showed that each DNA molecule conserve one strand from parental DNA that acts as a template for the formation of a new strand. So, DNA replicates semi-conservatively.

The complementary base pairing ensures the daughter DNA to have the same sequence of nucleotides as in the parent DNA. This specific relationship between bases is also known as Chargaff's rule. Erwin Chargaff is a scientist who had separated the four bases in DNA samples from various organisms. He observed that DNA has a 1:1 ratio of pyrimidine and purine bases and the number of A is about equal to T while the number of C is about equal to G. Chargaff's invention explains the basis of DNA replication.

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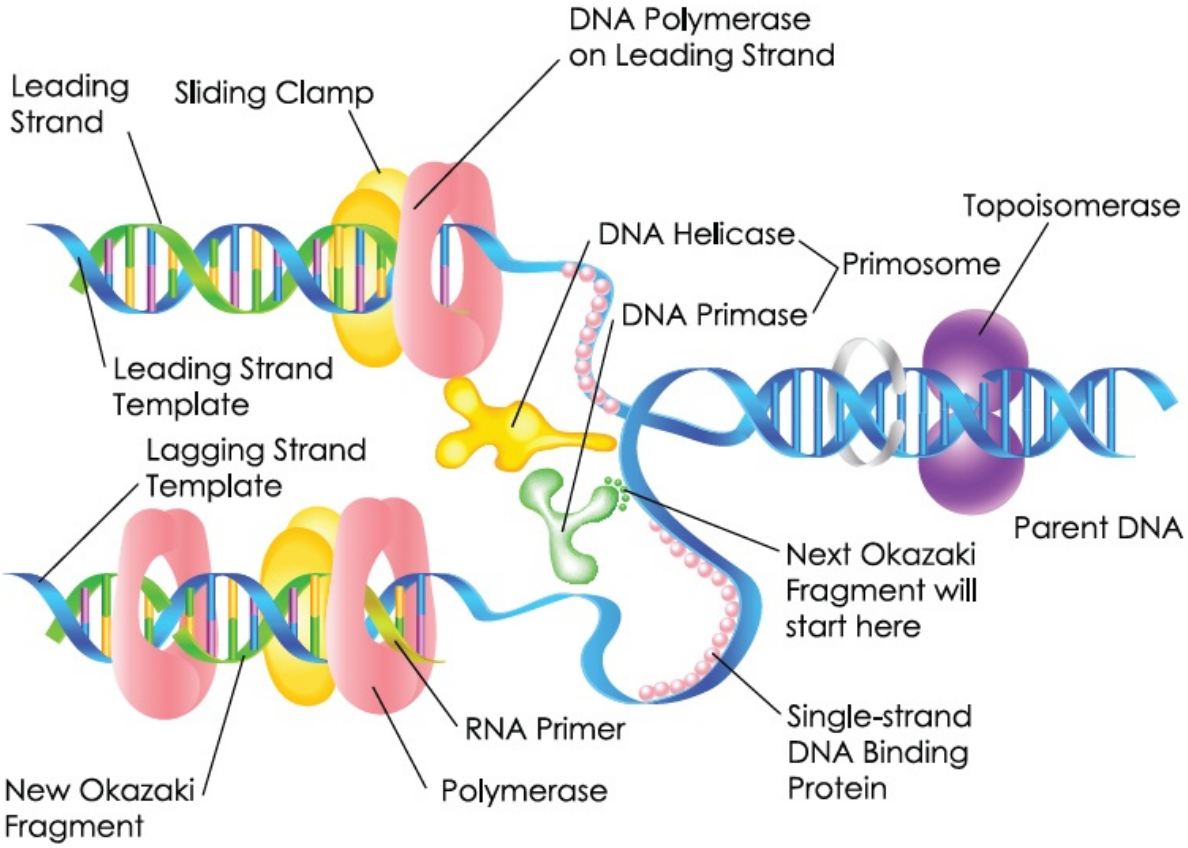
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DNA replication

DNA replication starts at a point called ori site (repilcation origin), and forms a Y shape-d structure called replication fork. This process involves several proteins. Helicase unwinds DNA strands to allow them to act as a template. Single-stranded binding (SSB) proteins bind to each strand stabilising them and avoiding from rejoining. Separation of DNA strands causes double helix to overwind ahead of a replication fork. Topoisomerase overwinds two DNA strands. RNA primers are starting points for synthesising new strands. They made by primase and binds to the 3’ end of the existing strand. DNA polymerase III synthesises new strand by adding nucleotides. It can add nucleotides from 3’ to 5’. One strand of parental DNA continuously builds new strand from 3’ to 5’ and known as leading strand. Second strand which runs in opposite direction from 5’ to 3’ known as lagging strand. Lagging strand is divided into sections called Okazaki fragments. RNA primers bind to the 3’ ends of each Okazaki fragment allowing DNA polymerase III to synthesise new strand. After each daughter strand built DNA polymerase I replaces RNA primers with DNA nucleotides. Ligase joins Okazaki fragments.

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Facts

- Telomeres are ends of chromosomes. They shorten in length with each replication, because of the primers used to synthesize the lagging strand. There is a limit of 52 DNA replications until the chromosomereaches the critical length. This limit is called the Hayflick limit.

Research time

Beside from semi-conservative model, there were two hypothesis for DNA replication: conservative and dispersive. Research each hypothesis and explain why they were rejected.

What results had to be shown in Meselson-Stahl experiment to prove these two hypothesis? Draw the samples for the first and second generation after replacement of 14N medium.

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Activity

a. Fill the table. Explain the function of each protein in replication using your own words. In a diagram section, draw the step of DNA replication involving each protein.

Protein	*Function*	*Diagram*
1.		
2.		
3.		

b. Bacteria divide about every 20 minutes. Bacteria cells were grown in a medium containing 15N. Then it was replaced on a medium containing 14N. After ultracentrifugation, first generation samples showed 50% of 15N and 50% of 14N. Samples were taken after 60 minutes too. What were the results in percentage?

Literacy

1. Describe the difference between the lagging and leading strands
2. To which group of atoms newly synthesised nucleotides join?

Terminology

- heavy - ауыр / тяжелый;
- lagging strand - кешігуші тізбек / отстающая нить;
- leading strand - жетекші тізбек / ведущая нить;
- medium - орта / среда;
- offspring - ұрпақ / отпрыск;
- replication fork - репликациялық айыр / репликационная вилка;
- sample - үлгі / образец;
- to ensure - қамтамасыз ету / обеспечивать;
- to inherit - тұқым қуалау / наследовать;
- to reject - қабылдамау / отклонить;

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- to replace - ауыстыру / заменять;
- to separate - бөлу / разделять;
- unwind - тарқату / раскручивать.

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1.13 RNA

You will:

- distinguish the structure and functions of RNA types.

Stimulating question

How does ribosome know which type of protein to make?

Key terms

- Ribosome - sphere-shaped organelle composed of protein and rRNA that serves as the site of protein synthesis;
- Messenger RNA - a type of RNA that convey genetic information from DNA to the ribosome, where they specify the amino acid sequence of the protein;
- Transfer RNA - a cloverleaf shaped RNA that transports specific amino acid to the ribosome;
- Ribosomal RNA - component of ribosome functioning in protein synthesis.

Text

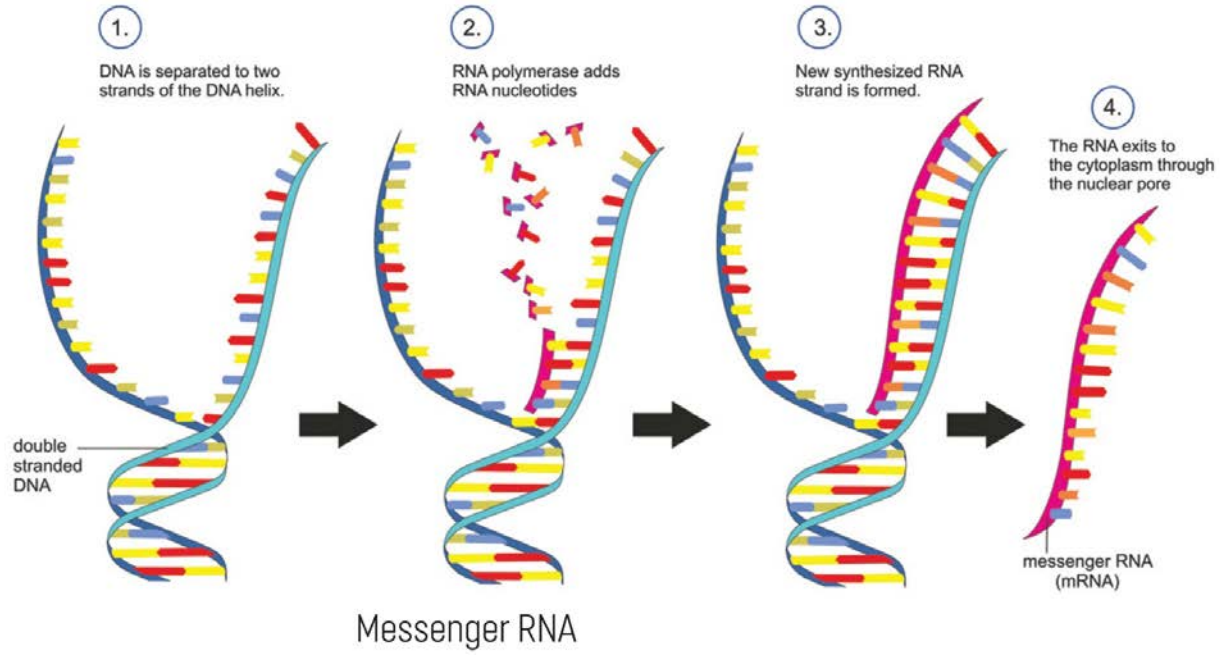
Ribonucleic acid or RNA is the type of nucleic acid. RNA consists of nucleotides. Nucleotide consists of three components: sugar, a phosphate group, and a nitrogenous base. RNA nucleotide includes pentose sugar (ribose) and phosphate group. As DNA, it contains adenine (A), cytosine (C), guanine (G), but instead of thymine it contains uracil (U). The main function of RNA is protein synthesis. RNA is involved in gene expression pathway by which genetic information in DNA is used to synthesize proteins that determine cell function. There are three types of RNA involved in protein synthesis: mRNA, tRNA, and rRNA.

Messenger RNA (mRNA)

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DNA dictates which protein will make cells. The sequence of bases in DNA serves as instructions for protein synthesis. Messenger RNA carries these instructions from DNA to a ribosome. mRNA is linear and single-stranded molecule. It is much shorter than DNA and contains about thousand nucleotides. DNA unwinds and unzips in particular regions, and one strand of DNA acts as a template for the synthesis of messenger RNA. Free RNA nucleotides complementary pairs with the bases of template DNA strand. Adenine attracts uracil and thymine attracts adenine while cytosine attracts guanine and guanine attract cytosine. RNA polymerase joins RNA nucleotides forming mRNA. Each sequence of three bases on the mRNA is called as codon and it codes for one amino acid.

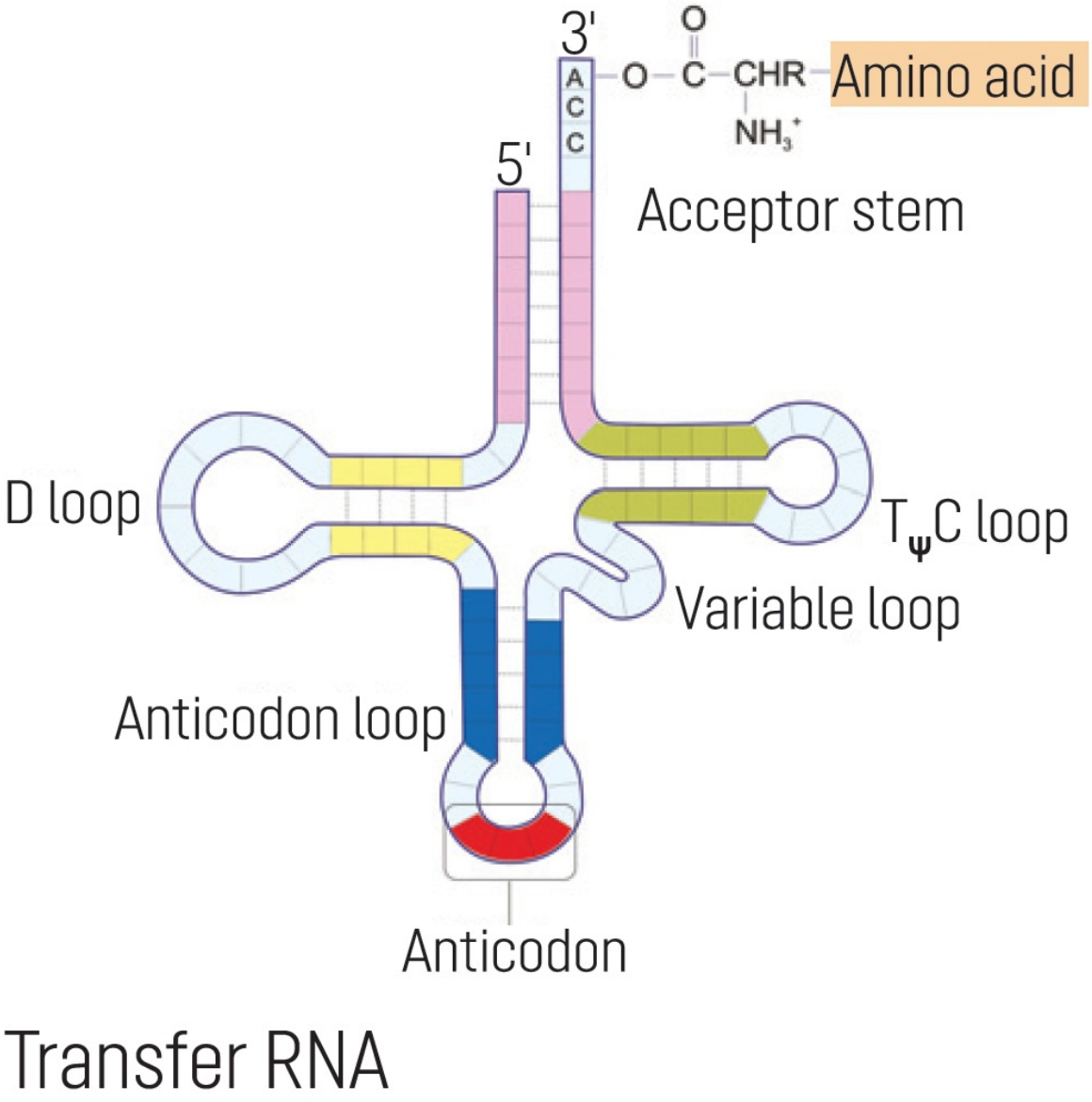


Transfer RNA (tRNA)

Ribosome requires amino acids for protein synthesis. Transfer RNA carries amino acids to the ribosome. They are small molecules containing about 75 nucleotides. The tRNA has folded structure with three loops that form a cloverleaf shape. One side of their structure binds to amino acid, and other side binds to mRNA. There are 20 types of tRNA according to the binding amino acid. Each tRNA binds only to specific amino acid. The side which attaches mRNA contains bases that are complementary to mRNA codon and known as anticodon.

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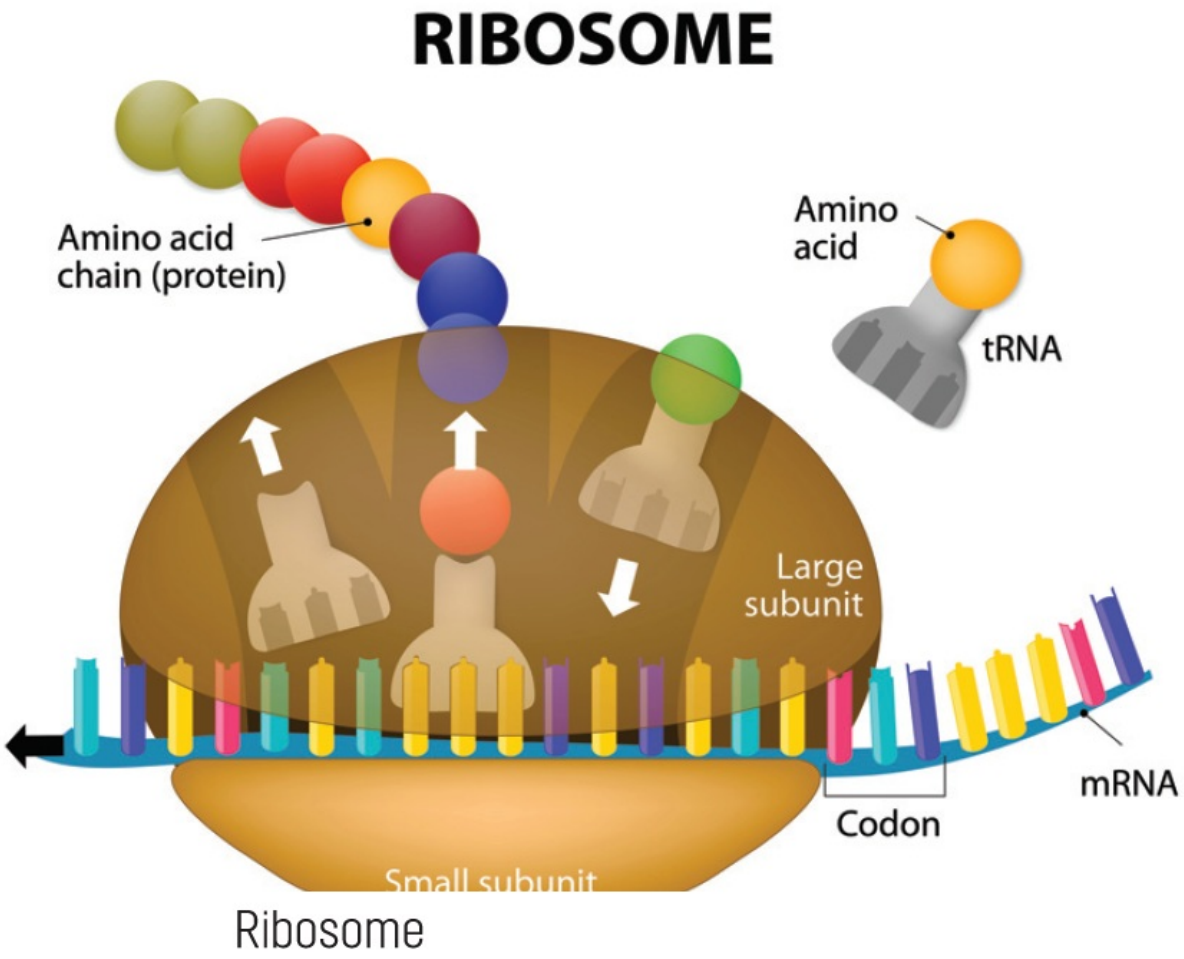


Ribosomal RNA (rRNA)

Ribosomal RNA combines with protein to form ribosomes. Ribosomes consist of two subunits: large and small. Both subunits contain rRNA. Ribosomes act as a site of protein synthesis and include enzymes involved in this process. They travel along the mRNA and read the information of the codons. tRNAs join to ribosomes and codon complementary binds with anticodon to synthesize a polypeptide chain.

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Research time

Protein synthesis includes two steps: transcription and translation. Give definitions and draw the process of protein synthesis involving RNA types.

Activity

a. Compare three types of RNA and fill the table. Based on features add a tick to the corresponding type of RNA.

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Features	*mRNA*	*tRNA*	*rRNA*
have anticodons			
join codon with an anticodon			
have codons			
assemble amino acids into a polypeptide chain			
have a cloverleaf shape			
component of the ribosome			
act as an intermediary between DNA and ribosomes			
contain uracil			
deliver amino acids			
single-stranded			
a version of the gene encoded by DNA			

b. The sequence of nucleotide in DNA strand is ACCTGAAGCTCT. What is the sequence of mRNA? What are the anticodons in tRNA?

c. There was a mutation in DNA strand, the second nucleotide is deleted. What is the sequence of mRNA? What are the anticodons in tRNA? Is protein in “b” the same with this protein? Explain your answer.

Literacy

1. How does mRNA leave the nucleus?
2. What is the scheme of the flow of genetic information?
3. DNA contains 50 nucleotides. How many amino acids will the resulting protein have?

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Terminology

- cloverleaf - беде жапырағы / клеверный лист;
- encoded - кодталған / закодированный;
- intermediary - делдал / посредник;
- messenger RNA - ақпараттық РНҚ / матричная РНК;
- ribosomal RNA - рибосомалық РНҚ / рибосомальная РНК;
- site - аумақ / участок;
- to assemble - жинау / собирать;
- to bind - байланыстыру / связать;
- to combine - біріктіру / объединять;
- to convey - тасымалдау / транспортировать;
- to deliver - жеткізу / доставлять;
- to distinguish - ажырату/ различать;
- to require - талап ету / требовать;
- transfer RNA - тасымалдаушы РНҚ / транспортная РНК.

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1.14 Nucleic acids

You will:

- compare the structure of DNA and RNA.

Stimulating question

Why genetic material is stored in DNA instead of RNA?

Key terms

- DNA - double-stranded nucleic acid that contains the genetic information;
- RNA - single-stranded nucleic acid that function in protein synthesis

Text

Nucleic acids are found in all living cells and in viruses. They perform vital functions in the cell and control the process of heredity and development.

Nucleic acids are organic compounds containing C, H, O, N, and P atoms. They are long polymers of units called nucleotides. Nucleotides are complex molecules comprising three components: sugar, a phosphate group, and a nitrogenous base. DNA and RNA are two types of nucleic acids. They are polynucleotides that differ in structure and function.

In both nucleic acids, the phosphate group of nucleotides include phosphoric acid.

However, pentose sugar in DNA is deoxyribose while in RNA is ribose. Deoxyribose differs from ribose that hydroxyl group at 2’ is replaced by a hydrogen atom.

There are five different nitrogenous bases found in nucleic acids, and they classified as purines and pyrimidines. Purines include adenine and guanine. They

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have double-ring structures whereas pyrimidines have a single ring structure. Pyrimidines involve thymine, cytosine, and uracil. DNA nucleotides contain adenine, guanine, cytosine, and thymine. In RNA nucleotide, thymine is replaced by uracil.

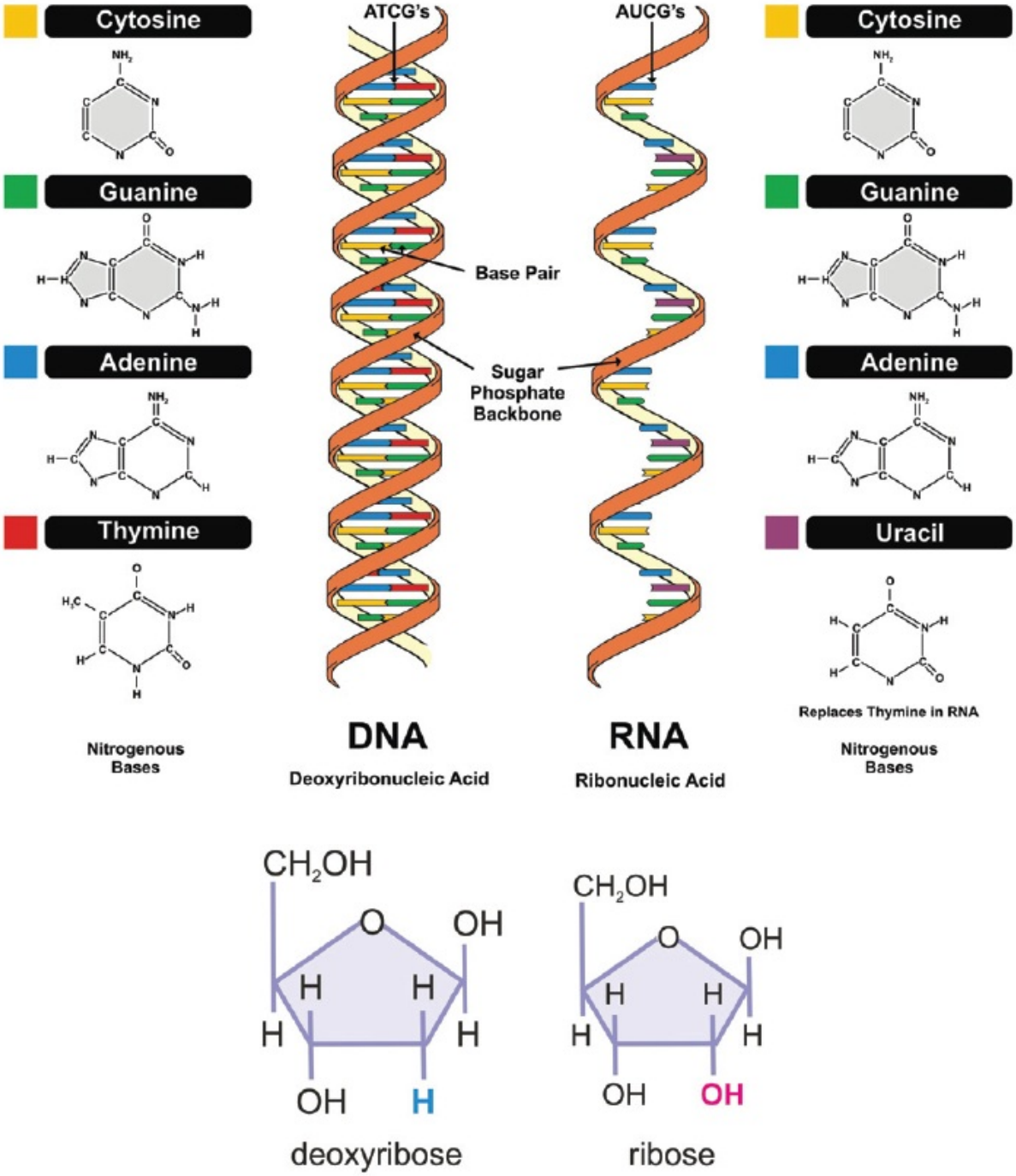
DNA is found in the nucleus, mitochondria, and chloroplast. It consists of two strands of long polynucleotide chains that form double helix structure. This structure makes DNA molecule very stable and ensures the storage of genetic information for years without any damage. Deoxyribose sugar

is less reactive because of C-H bonds. Also, DNA has small grooves that make damaging enzymes difficult to attach and attack it.

RNA contains only one polynucleotide chain. It is found in the cytoplasm, ribosome and nucleus. RNA can easily be altered by degradation and therefore it is unstable. Ribose sugar is more reactive due to the C-OH bonds. RNA has large grooves where enzymes can easily attach and attack the molecule.

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Research time

DNA is present in every cell of human body. Each DNA strand is 1.8 meters long but it is squeezed into a space of 0.09 micrometers. Research this phenomenon and explain how does DNA fit into the cell nucleus.

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Activity

a. Compare and contrast DNA and RNA. Find out their similarities and differences by drawing a Venn diagram.

Fact

If you unwrap all of the DNA you have in all your cells, you could reach the moon 6000 times.

Literacy

1. If you compare amino acid and nucleotide which one is more complicated? Explain your answer.
2. What is the Chargaff rule and which nucleic acid follows it?

Terminology

-
- groove - жырашық / канавка;
- heredity - тұқым қуалау / наследственность;
- ring - сақина / кольцо;
- to attach - бекіту / прикреплять;
- to comprise - қамту / включать;
- to damage - зақымдау / повредить;
- to ensure - қамтамасыз ету / обеспечивать;
- to fit - сыю / поместиться;
- to include - қамту / включать;
- to reach - жету / достигать;
- to replace - ауыстыру / заменять;
- to unwrap - жазу / развернуть;

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Problems

Test questions with one right answer

1. Fructose is:

A) ketose

B) disaccharide

C) aldose

D) triose

E) polysaccharide

2. The tendency of water molecules to stick together:

A) provides surface tension that allows insects to walk on water

B) acts to moderate temperature

C) keeps water moving through the vessels in a tree trunk

D) all of the above

E) none of the above

3. If a lipid is a solid at room temperature, you will predict that:

A) all lipids are solid at room temperature

B) the lipid is saturated

C) the lipid is unsaturated

D) all lipids are liquid at room temperature

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E) the lipid is a phospholipid

4. How many hydrogen bonds between cytosine and guanine?

A. 1

B. 2

C. 3

D. 4

E. 5

Test questions with several (max 3) right answers

1. Which of the followings are characteristics of saturated fats?

A) contain more hydrogen atoms

B) double bonds in hydrocarbon chain

C) single bonds in hydrocarbon chain

D) contain less hydrogen atoms

E) solid at room temperature

F) liquid at room temperature

G) mostly found in plants

H) have bends in hydrocarbon chain

2. Which of the followings are parts of amino acid?

A) −HN2

B) H2O

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C) −NH2

D) H

E) CO2

F) N2

G) −COH

H) −COOH

3. Which of the followings are pyrimidines?

A)adenine

B) ribose

C) pentose

D) guanine

E) cytosine

F) uracil

G) deoxyribose

H) thymine

Matching

1. Match the following type of bonds with its molecules.

1. Glycosidic linkage

2. Ester bond

3. Peptide bond

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A) between water molecules

B) between amino acids

C) between nucleotides

D) between monosaccharides

E) between glycerol and fatty acid

F) between fatty acids

2. Match the following proteins with their prosthetic groups.

1. Chromoproteins

2. Phosphoproteins

3. Glycoproteins

A) lipids

B) retinal

C) sodium group

D) carbohydrate

E) phosphate group

F) nucleotide

*Книга предоставлена исключительно в образовательных целях
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